Prepared for submission to JHEP

SOFTSUSY: a program for calculating supersymmetric spectra

B.C. Allanach[a]

[a] *DAMTP, CMS, University of Cambridge, Wilberforce road, Cambridge, CB3 0WA, United Kingdom*

E-mail: B.C.Allanach@damtp.cam.ac.uk

Abstract: SOFTSUSY is a program which accurately calculates the spectrum of superparticles in the CP-conserving Minimal Supersymmetric Standard Model (MSSM), with a full flavour mixing structure. The program solves the renormalisation group equations with theoretical constraints on soft supersymmetry breaking terms provided by the user. Weak-scale gauge coupling and fermion mass data (including one-loop finite MSSM corrections) are used as a boundary condition, as well as successful radiative electroweak symmetry breaking. The program can also calculate a measure of fine-tuning. The program structure has been designed to easily generalise to extensions of the MSSM. This article serves as a self-contained guide to prospective users, and indicates the conventions and approximations used.

Keywords: sparticle, MSSM

Contents

1 Introduction **2**
- 1.1 The Nature of the Physical Problem 2
- 1.2 The Program 2
- 1.3 Aims and Layout 3

2 MSSM Parameters **3**
- 2.1 Supersymmetric Parameters 3
- 2.2 SUSY Breaking Parameters 4
- 2.3 Tree-Level Masses 4

3 Calculation **6**
- 3.1 Below $M_{QED\times QCD5}$ 6
- 3.2 Initial Estimate 7
- 3.3 Gauge and Yukawa Couplings 7
- 3.4 MSSM Renormalisation 9
- 3.5 Electroweak Symmetry Breaking 10
- 3.6 MSSM Spectrum 11
- 3.7 Fine Tuning 12

A Running SOFTSUSY **12**
- A.1 Input file 13

B Sample Program **15**

C Sample Output **18**
- C.1 Problem flags 19

D Switches and Constants **21**

E Object Structure **21**
- E.1 Linear Algebra 21
- E.2 General Structure 22
- E.3 `RGE` Class 22
- E.4 `Approx` Class 23
- E.5 `QedQcd` Class 23
- E.6 `MssmSusy` Class 24
- E.7 `SoftParsMssm` Class 24
- E.8 `MssmSoftsusy` Class 25
- E.9 `FlavourMssmSoftsusy` Class 29

1 Introduction

The Minimal Supersymmetric Standard Model (MSSM) provides an attractive weak-scale extension to the Standard Model. As well as solving the gauge hierarchy problem, it can be motivated by more fundamental models such as various string theories or supersymmetric grand unified theories. The MSSM provides a rich and complicated phenomenology. It predicts many states extra to the Standard Model (sparticles) and their indirect empirical effects and direct detection are vital for verification of the MSSM. Models that are more fundamental than the MSSM can provide stringent constraints upon the way supersymmetry (SUSY) is broken, with important implications for the spectrum which in turn affects the signatures available in experiments. It is therefore desirable to construct a calculational tool which may provide a spectrum and couplings of the MSSM sparticles so that studies of the capabilities of colliders, extraction of high scale parameters (if a signal is observed) and studies of constraints on the models are enabled. We present such a tool (`SOFTSUSY`) in this article.

1.1 The Nature of the Physical Problem

The determination of sparticle masses and couplings of SUSY particles in the R-parity conserving MSSM is the basic problem. Low energy data on Standard Model fermion masses, gauge couplings and electroweak boson masses are to be used as a constraint. SUSY radiative corrections to these inputs from sparticle loops depend upon the sparticle spectrum, and must be calculated. Theoretical constraints on the SUSY breaking parameters from an underlying theory are often imposed at a high renormalisation scale, perhaps resulting from a super gravity or string theory. Often, the theoretical constraints drastically reduce the number of free parameters in the SUSY breaking sector (which numbers over 100 in the unconstrained case). These constraints then make phenomenological analysis tractable by reducing the dimensionality of parameter space sufficiently so that parameter scans over a significant volume of parameter space are possible. Finally, the MSSM parameters must also be consistent with a minimum in the Higgs potential which leads to the observed electroweak boson masses.

This problem has been addressed many times before in the literature (see for example [1–5]), with varying degrees of accuracy in each part of the calculation. It is our purpose here to provide a tool which will solve the problem with a high accuracy, including state-of-the-art corrections. Similar problems in the context of MSSM extensions[1] have also been studied. In anticipation of new forms of SUSY breaking constraints and new MSSM extensions, we designed the tool to be flexible and easily extended.

1.2 The Program

`SOFTSUSY` has been written in object-oriented C++ but users may use an executable program with input either in the SUSY Les Houches Accord [30] format or from command-line arguments. For users wishing to call `SOFTSUSY` from their own programs, the user interface is designed to be C-like to aid users that are unfamiliar with object orientation. Accuracy and ease of generalisation have taken priority over running speed in the design. For example, full three family mass and Yukawa matrices may be employed, rather than the more usual dominant third family approximation. The publicly released codes `ISASUGRA` (which comprises part of the `ISAJET` package [5]) and `SUSPECT` [4] use the dominant third family approximation, for example. The full three-family choice slows the renormalisation group evolution significantly, but will facilitate studies of sparticle or quark mixing.

[1] By MSSM extension, we mean an extension applicable near the weak scale.

The running time is not foreseen as a bottleneck because it is a matter of a couple of seconds on a modern PC, and will certainly be negligible compared to any Monte-Carlo simulation of sparticle production and decay in colliders. It is possible for the user to specify their own high scale boundary conditions for the soft SUSY breaking parameters without having to change the SOFTSUSY code. For the convenience of most users however, the most commonly used high-scale boundary conditions are included in the package.

The code can be freely obtained from the SOFTSUSY web-page, which currently resides at the address

https://ballanach.github.io/softsusy/

Installation instructions and more detailed technical documentation of the code may also be found there.

SOFTSUSY is a tool whose output could be used for studies of MSSM sparticle searches [6] by using event generators such as HERWIG [7], or other more theoretical or astrophysical studies. For a review of SUSY tools on offer (which may use the output from SOFTSUSY), see Ref. [8].

1.3 Aims and Layout

The main aims of this article are to provide a manual for the use of SOFTSUSY, to describe the approximations employed and to detail the notation used in order to allow for user generalisation. There have been other articles published on the comparison of the calculation in SOFTSUSY with those of other codes [9], and so we decline from including such information here.

The rest of this paper proceeds as follows: the relevant MSSM parameters are presented in sec. 2. The approximations employed are noted in sec. 3, but brevity requires that they are not explicit. However, a reference is given so that the precise formulae utilised may be obtained in each case. The algorithm of the calculation is also outlined. Technical information related to running and extending the program is placed in appendices. A description of how to run the command-line interface is given in appendix A, including information on the input-file. Appendix B gives an example of a main program, useful if the user wants to call SOFTSUSY from his or her main program. The sample output from this program is displayed and explained in appendix C. The use of switches and constants is explained in appendix D. Finally, in appendix E, a description of the relevant objects and their relation to each other is presented.

2 MSSM Parameters

In this section, we introduce the MSSM parameters in the SOFTSUSY conventions. Translations to the actual variable names used in the source code are shown in appendix E.

2.1 Supersymmetric Parameters

The chiral superfields of the MSSM have the following $G_{SM} = SU(3)_c \times SU(2)_L \times U(1)_Y$ quantum numbers

$$\begin{aligned} &L : (1, 2, -\tfrac{1}{2}), \quad \bar{E} : (1, 1, 1), \qquad Q : (3, 2, \frac{1}{6}), \quad \bar{U} : (\bar{3}, 1, -\frac{2}{3}), \\ &\bar{D} : (\bar{3}, 1, \tfrac{1}{3}), \quad H_1 : (1, 2, -\frac{1}{2}), \quad H_2 : (1, 2, \frac{1}{2}). \end{aligned} \tag{2.1}$$

Then, the superpotential is written as

$$W = \epsilon_{ab}\left[(Y_E)_{ij}L_i^b H_1^a \bar{E}_j + (Y_D)_{ij}Q_i^{bx}H_1^a\bar{D}_{jx} + (Y_U)_{ij}Q_i^{ax}H_2^b\bar{U}_{jx} + \mu H_1^b H_2^a\right] \tag{2.2}$$

Throughout this section, we denote an $SU(3)$ colour index of the fundamental representation by $x, y, z = 1, 2, 3$. The $SU(2)_L$ fundamental representation indices are denoted by $a, b, c = 1, 2$ and the generation indices by $i, j, k = 1, 2, 3$. $\epsilon_{ab} = \epsilon^{ab}$ is the totally antisymmetric tensor, with $\epsilon_{12} = 1$. Note that the sign of μ is identical to the one in `ISASUGRA` [5], but is in the opposite convention to ref. [3]. Presently, real Yukawa couplings only are included. All MSSM running parameters are in the $\overline{DR}$ scheme. The Higgs vacuum expectation values (VEVs) are $\langle H_i^0\rangle = v_i/\sqrt{2}$ and $\tan\beta = v_2/v_1$. g_i are the MSSM $\overline{DR}$ gauge couplings and g_1 is defined in the Grand Unified normalisation $g_1 = \sqrt{5/3}g'$, where g' is the Standard Model hypercharge gauge coupling. Elements of fermion mass matrices are given by

$$(m_u)_{ij} = \frac{1}{\sqrt{2}}(Y_U)_{ij}v_2, \qquad (m_{d,e})_{ij} = \frac{1}{\sqrt{2}}(Y_{D,E})_{ij}v_1 \tag{2.3}$$

for the up quark, down quark and charged lepton matrices respectively.

2.2 SUSY Breaking Parameters

We now tabulate the notation of the soft SUSY breaking parameters. The trilinear scalar interaction potential is

$$V_3 = \epsilon_{ab}\left[\tilde{Q}_{i_L}^{xa}(U_A)_{ij}\tilde{u}_{jx_R}H_2^b + \tilde{Q}_{i_L}^{xb}(D_A)_{ij}\tilde{d}_{jx_R}H_1^a + \tilde{L}_{i_L}^b(E_A)_{ij}\tilde{e}_{j_R}H_1^a + H.c.\right], \tag{2.4}$$

where fields with a tilde are the scalar components of the superfield with the identical capital letter. Also,

$$(A_{U,D,E})_{ij} = (U_A, D_A, E_A)_{ij}/(Y_{U,D,E})_{ij} \tag{2.5}$$

(no summation on i, j) are often referred to in the literature as soft A-parameters.

The scalar bilinear SUSY breaking terms are contained in the potential

$$\begin{aligned} V_2 = {} & m_{H_1}^2 {H_1}_a{}^* H_1^a + m_{H_2}^2 {H_2}_a{}^* H_2^a + \tilde{Q}_{ixa}^*(m_{\tilde{Q}}^2)_{ij}\tilde{Q}_j^{xa} + \tilde{L}_{ia}^*(m_{\tilde{L}}^2)_{ij}\tilde{L}_j^a + \\ & \tilde{u}_i^x(m_{\tilde{u}}^2)_{ij}\tilde{u}_{jx}^* + \tilde{d}_i^x(m_{\tilde{d}}^2)_{ij}\tilde{d}_{jx}^* + \tilde{e}_i(m_{\tilde{e}}^2)_{ij}\tilde{e}_j^* + \epsilon_{ab}(m_3^2 H_2^a H_1^b + H.c.). \end{aligned} \tag{2.6}$$

For a comparison of these conventions with other popular ones in the literature, see Table 1. In the table, we compare the `SOFTSUSY` conventions with the SUSY Les Houches Accord [30] and ref. [32] (Martin and Vaughn).

Writing the bino as $\tilde{b}$, $\tilde{w}^{A=1,2,3}$ as the unbroken-SU(2)$_L$ gauginos and $\tilde{g}^{X=1\ldots8}$ as the gluinos, the gaugino mass terms are contained in the Lagrangian

$$\mathcal{L}_G = \frac{1}{2}\left(M_1\tilde{b}\tilde{b} + M_2\tilde{w}^A\tilde{w}^A + M_3\tilde{g}^X\tilde{g}^X\right) + h.c. \tag{2.7}$$

2.3 Tree-Level Masses

Here we suppress any gauge indices and follow the notation of ref. [3] closely. The Lagrangian contains the neutralino mass matrix as $-\frac{1}{2}\tilde{\psi}^{0T}\mathcal{M}_{\tilde{\psi}^0}\tilde{\psi}^0$ + h.c., where $\tilde{\psi}^0 = (-i\tilde{b}, -i\tilde{w}^3, \tilde{h}_1, \tilde{h}_2)^T$ and

$$\mathcal{M}_{\tilde{\psi}^0} = \begin{pmatrix} M_1 & 0 & -M_Z c_\beta s_W & M_Z s_\beta s_W \\ 0 & M_2 & M_Z c_\beta c_W & -M_Z s_\beta c_W \\ -M_Z c_\beta s_W & M_Z c_\beta c_W & 0 & -\mu \\ M_Z s_\beta s_W & -M_Z s_\beta c_W & -\mu & 0 \end{pmatrix}. \tag{2.8}$$

SOFTSUSY	SLHA	Martin and Vaughn
Y^U	Y^U	$(Y^U)^T$
$Y^{D,E}$	$Y^{D,E}$	$(-Y^{D,E})^T$
U_A	T_U	h_U^T
D_A, E_A	$T_{D,E}$	$(-h_{D,E})^T$
$m^2_{\tilde{Q},\tilde{L}}$	$m^2_{\tilde{Q},\tilde{L}}{}^T$	$m^2_{\tilde{Q},\tilde{L}}$
$m^2_{\tilde{u},\tilde{d},\tilde{e}}$	$m^2_{\tilde{u},\tilde{d},\tilde{e}}$	$m^2_{\tilde{u},\tilde{d},\tilde{e}}$
μ	μ	μ
m_3^2	$B\mu$	B
M_i	M_i	M_i
$m^2_{H_{1,2}}$	$m^2_{H_{1,2}}$	$m^2_{H_{d,u}}$

Table 1. Comparison of conventions between SOFTSUSY and the literature. Note that simple models of SUSY breaking, for example the CMSSM, will have negative gaugino masses M_i in the SOFTSUSY conventions. The sign in front of the gaugino mass term in front of the SOFTSUSY Lagrangian appears at first sight to be in the opposite convention to Ref. [32]. However, it is due to the absorption of a factor of i into the gaugino fields $\tilde{B}$, $\tilde{W}$ and $\tilde{g}$ which means that the sign of the gaugino mass is identical to that of [32], and is thus of the same convention.

We use s and c for sine and cosine, so that $s_\beta \equiv \sin\beta$, $c_\beta \equiv \cos\beta$ and $s_W(c_W)$ is the sine (cosine) of the weak mixing angle. The 4 by 4 neutralino mixing matrix is an orthogonal matrix O with real entries, such that $O^T \mathcal{M}_{\tilde{\psi}^0} O$ is diagonal. The neutralinos χ_i^0 are defined such that their absolute masses increase with increasing i. Some of their mass values can be negative.

We make the identification $\tilde{w}^\pm = (\tilde{w}^1 \mp i\tilde{w}^2)/\sqrt{2}$ for the charged winos and $\tilde{h}_1^-, \tilde{h}_2^+$ for the charged higgsinos. The Lagrangian contains the chargino mass matrix as $-\tilde{\psi}^{-T}\mathcal{M}_{\tilde{\psi}^+}\tilde{\psi}^+ + h.c.$, where $\tilde{\psi}^+ = (-i\tilde{w}^+,\ \tilde{h}_2^+)^T$, $\tilde{\psi}^- = (-i\tilde{w}^-,\ \tilde{h}_1^-)^T$ and

$$\mathcal{M}_{\tilde{\psi}^+} \ = \ \begin{pmatrix} M_2 & \sqrt{2}\, M_W s_\beta \\ \sqrt{2}\, M_W c_\beta & \mu \end{pmatrix}. \tag{2.9}$$

This matrix is then diagonalised by 2 dimensional rotations through angles θ_L, θ_R in the following manner:

$$\begin{pmatrix} c_{\theta_L} & s_{\theta_L} \\ -s_{\theta_L} & c_{\theta_L} \end{pmatrix} \mathcal{M}_{\tilde{\psi}^+} \begin{pmatrix} c_{\theta_R} & -s_{\theta_R} \\ s_{\theta_R} & c_{\theta_R} \end{pmatrix} = \begin{pmatrix} m^+_{\chi_1} & 0 \\ 0 & m^+_{\chi_2} \end{pmatrix} \tag{2.10}$$

where $m^+_{\chi_i}$ could be negative, with the mass parameter of the lightest chargino being in the top left hand corner.

At tree level the gluino mass, $m_{\tilde{g}}$, is given by M_3.

Strong upper bounds upon the inter-generational scalar mixing exist [10] and in the following we assume that such mixings are negligible. The tree-level squark and slepton mass squared values for the family i are found by diagonalising the following mass matrices $\mathcal{M}^2_{\tilde{f}}$ defined in the $(\tilde{f}_{iL}, \tilde{f}_{iR})^T$ basis:

$$\begin{pmatrix} (m^2_{\tilde{Q}})_{ii} + m^2_{u_i} + (\frac{1}{2} - \frac{2}{3}s^2_W)M^2_Z c_{2\beta} & m_{u_i}\left((A_U)_{ii} - \mu\cot\beta\right) \\ m_{u_i}\left((A_U)_{ii} - \mu\cot\beta\right) & (m^2_{\tilde{u}})_{ii} + m^2_{u_i} + \frac{2}{3}s^2_W M^2_Z c_{2\beta} \end{pmatrix}, \tag{2.11}$$

$$\begin{pmatrix} (m^2_{\tilde{Q}})_{ii} + m^2_{d_i} - (\frac{1}{2} - \frac{1}{3}s^2_W)M^2_Z c_{2\beta} & m_{d_i}\left((A_D)_{ii} - \mu\tan\beta\right) \\ m_{d_i}\left((A_D)_{ii} - \mu\tan\beta\right) & (m^2_{\tilde{d}})_{ii} + m^2_{d_i} - \frac{1}{3}s^2_W M^2_Z c_{2\beta} \end{pmatrix}, \tag{2.12}$$

$$\begin{pmatrix} (m^2_{\tilde{L}})_{ii} + m^2_{e_i} - (\frac{1}{2} - s^2_W)M^2_Z c_{2\beta} & m_{e_i}\left((A_E)_{ii} - \mu\tan\beta\right) \\ m_{e_i}\left((A_E)_{ii} - \mu\tan\beta\right) & (m^2_{\tilde{e}})_{ii} + m^2_{e_i} - s^2_W M^2_Z c_{2\beta} \end{pmatrix}, \tag{2.13}$$

m_f, e_f are the mass and electric charge of fermion f respectively. The mixing of the first two families is suppressed by a small fermion mass, which we approximate to zero. The non-sneutrino sfermion mass eigenstates are given by

$$\begin{pmatrix} m^2_{\tilde{f}_1} & 0 \\ 0 & m^2_{\tilde{f}_2} \end{pmatrix} = \begin{pmatrix} c_f & s_f \\ -s_f & c_f \end{pmatrix} \mathcal{M}^2_{\tilde{f}} \begin{pmatrix} c_f & -s_f \\ s_f & c_f \end{pmatrix} \tag{2.14}$$

where c_f is the cosine of the sfermion mixing angle, $\cos\theta_f$, and s_f the sine. θ_f are set in the convention that the two mass eigenstates are in no particular order and $\theta_f \in [-\pi/4, \pi/4]$. The sneutrinos have masses corresponding to

$$m^2_{\tilde{\nu}_i} = (m^2_{\tilde{L}})_{ii} + \frac{1}{2}M^2_Z c_{2\beta}. \tag{2.15}$$

The CP-even gauge eigenstates (H^0_1, H^0_2) are rotated by the angle α into the mass eigenstates ($H^0\, h^0$) as follows,

$$\begin{pmatrix} H^0 \\ h^0 \end{pmatrix} = \begin{pmatrix} c_\alpha & s_\alpha \\ -s_\alpha & c_\alpha \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\Re H^0_1 \\ \frac{1}{\sqrt{2}}\Re H^0_2 \end{pmatrix}. \tag{2.16}$$

$m_{h^0} < m_{H^0}$ by definition, and $\alpha \in [-\pi/4, 3\pi/4]$. The CP-odd and charged Higgs masses are

$$m^2_{A^0} = m^2_3(\tan\beta + \cot\beta), \qquad m^2_{H^\pm} = m^2_{A^0} + M^2_W \tag{2.17}$$

at tree level.

3 Calculation

We now show the algorithm used to perform the calculation. Standard Model parameters (fermion and gauge boson masses, the fine structure constant $\alpha(M_Z)$, the Fermi constant from muon decay G^μ_F and $\alpha_3(M_Z)$) are used as constraints. The soft SUSY breaking parameters and the superpotential parameter μ are then the free parameters. However, in what follows, $|\mu|$ is constrained by M_Z and $\tan\beta$ is traded for m_3 as an input parameter. Therefore, the total list of unconstrained input parameters is: any fundamental soft SUSY breaking parameters (except m^2_3), $\tan\beta$ and the sign of μ. First we describe the evolution of the low-energy Standard Model input parameters below a scale where 5 quark flavour QED×QCD is matched onto the MSSM ($M_{QED\times QCD5}$), then detail the rest of the algorithm.

3.1 Below $M_{QED\times QCD5}$

$\alpha(M_Z)$, $\alpha_s(M_Z)$ are first evolved to 1 GeV using 3 loop QCD and 1 loop QED [11–13] with step-function decoupling of fermions at their running masses. We have checked that the contribution from 2-loop matching [14] is negligible; the effect of 3-loop terms in the renormalisation group equations is an order of magnitude larger. Then, the two gauge couplings and all Standard Model fermion masses except the top mass are run to $M_{QED\times QCD5}$. The β functions of fermion masses are taken to be zero at renormalisation scales below their running masses. The parameters at $M_{QED\times QCD5}$ are used as the low energy boundary condition in the rest of the evolution.

3.2 Initial Estimate

The algorithm proceeds via the fixed point iteration method, and therefore an approximate initial guess of MSSM parameters is required. For this, the third family $\overline{DR}$ Yukawa couplings are approximated by

$$h_t(Q) = \frac{m_t(Q)\sqrt{2}}{v \sin\beta}, \qquad h_{b,\tau}(Q) = \frac{m_{b,\tau}(Q)\sqrt{2}}{v \cos\beta}, \tag{3.1}$$

where $v = 246.22$ GeV is the Standard Model Higgs VEV and $Q = M_{QED\times QCD5}$ is the renormalisation scale. The $\overline{MS}$ values of fermion masses are used for this initial estimate. The fermion masses and α_s at the top mass are obtained by evolving the previously obtained fermion masses and gauge couplings from $M_{QED\times QCD5}$ to m_t (with the same accuracy). The electroweak gauge couplings are estimated by $\alpha_1(M_{QED\times QCD5}) = 5\alpha(M_{QED\times QCD5})/(3c_W^2)$, $\alpha_2(M_{QED\times QCD5}) = \alpha(M_{QED\times QCD5})/s_W^2$. Here, s_W is taken to be the on-shell value. These two gauge couplings are then evolved to m_t with 1-loop Standard Model β functions, including the effect of a light higgs (without decoupling it). In this initial guess, no SUSY threshold effects are calculated. The gauge and Yukawa couplings are then evolved to the unification scale M_X with the one-loop MSSM β functions, where the user-supplied boundary condition on the soft terms is applied. Also, $\mu(M_X) = \mathrm{sgn}(\mu) \times 1$ GeV and $m_3(M_X) = 0$ are imposed. These initial values are irrelevant; they are overwritten on the next iteration by more realistic boundary conditions. $\mu(M_X)$ is set to be of the correct sign because its sign does not change through renormalisation.

The whole system of MSSM soft parameters and SUSY couplings is then evolved to 1-loop order to $M_{QED\times QCD5}$. At $M_{QED\times QCD5}$, the tree-level electroweak symmetry breaking (EWSB) conditions are applied [6] to predict μ and m_3. The masses and mixings of MSSM super particles are then calculated at tree-level order by using the SUSY parameters (and m_3) calculated at $M_{QED\times QCD5}$. The resulting set of MSSM parameters is then used as the initial guess for the iterative procedure described below.

3.3 Gauge and Yukawa Couplings

Figure 3.2 shows the iterative procedure, starting from the the top. The whole calculation is currently performed in the real full three family approximation, i.e. all Yukawa couplings are set to be real, but quark mixing is incorporated. First of all, the one-loop radiative corrections are applied to the gauge and third-family Yukawa couplings. For these, we rely heavily on ref. [3] by Bagger, Matchev, Pierce and Zhang (BMPZ)[2]. In the threshold corrections, we use running $\overline{DR}$ masses and parameters at the relevant scale, unless denoted otherwise in the text. $m_t(M_{QED\times QCD5})$ is calculated with 2-loop QCD [15] and the full one-loop supersymmetric contributions to $m_t(M_{QED\times QCD5})$, including logarithmic and finite contributions (Σ_t^{BMPZ}, eqs. (D.16)-(D.18) of BMPZ):

$$m_t(M_{QED\times QCD5})^{\overline{DR}}_{MSSM} = \Sigma_t^{BMPZ} + m_t^{pole}. \tag{3.2}$$

These corrections are important because the region of valid EWSB is very sensitive to m_t [6]. To calculate $m_b(M_{QED\times QCD5})^{\overline{DR}}$, we first calculate the Standard Model $\overline{DR}$ value from the $\overline{MS}$ one [15, 33]

$$m_b(M_{QED\times QCD5})^{\overline{DR}}_{SM} = m_b(M_{QED\times QCD5})^{\overline{MS}}_{SM}\left(1 - \frac{\alpha_s}{3\pi} + \frac{3g^2}{128\pi^2} + \frac{13g_1^2}{1152\pi^2}\right). \tag{3.3}$$

We then add the leading one-loop supersymmetric corrections

$$m_b(M_{QED\times QCD5})^{\overline{DR}}_{MSSM} = m_b(M_{QED\times QCD5})^{\overline{DR}}_{SM}/(1 + \Delta^b_{SUSY}). \tag{3.4}$$

[2]Whenever a reference to an equation in BMPZ is made, it is understood that the sign of μ must be reversed.

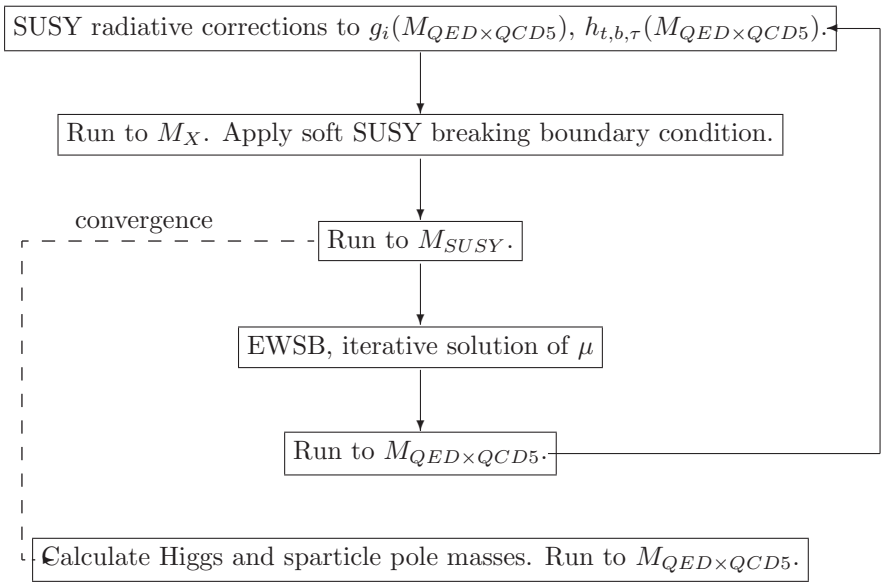


Figure 1. Fixed point iteration algorithm used to calculate the SUSY spectrum. Each step (represented by a box) is detailed in the text. The initial step is the uppermost one. M_{SUSY} is defined to be the scale at which the EWSB conditions are imposed, as discussed in the text. M_X is the scale at which the high energy SUSY breaking boundary conditions are imposed.

The contributions to Δ^b_{SUSY} are included in full from eq. D.18 of BMPZ (neglecting the term proportional to e, since that is already included in the QED calculation of the SM $m_b(M_{QED\times QCD5})$). Both finite and leading logarithmic corrections are included. After the Standard Model $\overline{MS}$ value of m_τ is converted to the $\overline{DR}$ value via

$$m_\tau(M_{QED\times QCD5})^{\overline{DR}}_{SM} = m_\tau(M_{QED\times QCD5})^{\overline{MS}}_{SM}\left(1 - \frac{3}{128\pi^2}(g_1^2 - g_2^2)\right). \tag{3.5}$$

The full one-loop MSSM corrections from the appendix of BPMZ (aside from the photon contribution, since that has already been included in $m_\tau(M_{QED\times QCD5})^{\overline{MS}}_{SM}$) are then used to correct $m_\tau(M_{QED\times QCD5})^{\overline{DR}}_{SM}$:

$$m_\tau(M_{QED\times QCD5})^{\overline{DR}}_{MSSM} = m_\tau(M_{QED\times QCD5})^{\overline{DR}}_{SM}(1 + \Sigma_\tau). \tag{3.6}$$

The one-loop $\overline{DR}$ values for $m_t(M_{QED\times QCD5})$, $m_b(M_{QED\times QCD5})$, $m_\tau(M_{QED\times QCD5})$ are then substituted with the $\overline{DR}$ value of $v(Q)$ into eq. (2.3) to calculate the third family $\overline{DR}$ Yukawa couplings at $M_{QED\times QCD5}$. $v(Q)$ is run to two-loops. The other diagonal elements of the Yukawa matrices are set by eq. (2.3) but with fermion masses replaced by the $\overline{MS}$ values.

The default option is to perform the calculation in the dominant third-family approximation, where all elements of Yukawa matrices expect for the (3,3) elements are set to zero. There are also options described in appendix D for performing the calculation in the unmixed 3-family approximation or the fully-mixed 3-family case. If a flavour-mixing option is chosen (see section E.9), the Yukawa couplings are then mixed using the "standard parameterisation" of the CKM matrix [17] with CP-violating phase set either to zero or π, whichever results in a positive entry for $(V_{CKM})_{13}$ (also known

as V_{ub}):

$$V_{CKM} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & ps_{13} \\ -s_{12}c_{23} - pc_{12}s_{23}s_{13} & c_{12}c_{23} - ps_{12}s_{23}s_{13} & s_{23}c_{13} \\ s_{12}s_{23} - pc_{12}c_{23}s_{13} & -c_{12}s_{23} - ps_{12}c_{23}s_{13} & c_{23}c_{13} \end{pmatrix}, \tag{3.7}$$

where $s_{ij} \equiv \sin\theta_{ij}$, $c_{ij} \equiv \cos\theta_{ij}$ and $p = \pm 1$. Sign conventions are automatically chosen such that diagonal entries and entries above the diagonal in V_{CKM} are positive. Note that V_{CKM} is a member of $O(3)$, i.e. $V_{CKM}^{-1} = V_{CKM}^{T}$. While complex phase effects are obviously not taken into account in eq. 3.7, it is hoped that the magnitudes of the main quark mixing effects will be. In fact, using the central values for $|(V_{CKM})_{ij}|$ given by the particle data group [17], the magnitudes of all elements in V_{CKM} in the first row and last column may be exactly reproduced. Of the other entries, $|V_{cs}|$ is accurate at the 2×10^{-5} level and $|V_{cd}|$ at the 4×10^{-4} level: surely negligible for most practical purposes. $|V_{ts}|$ is also quite accurate (to 1.2%), but it should be noted that $|V_{td}|$ is wrong by around 50%. Any flavour physics effects sensitive to $|V_{td}|$ is therefore subject to this large uncertainty on its value issuing from SOFTSUSY[3]. The up (by default), or down Yukawa couplings at $M_{QED\times QCD5}$ are mixed in the weak eigenbasis via

$$(Y_U)' = V_{CKM}^{T}(Y^U)V_{CKM}, \qquad (Y_D)' = V_{CKM}(Y^D)V_{CKM}^{T} \tag{3.8}$$

where the primed Yukawa matrix is in the weak eigenbasis and the unprimed is in the mass eigenbasis.

Full one-loop corrections to $g_i(M_{QED\times QCD5})$ are included. The two-loop treatment of electroweak gauge couplings follows from appendix C of BMPZ. The pole value of m_t is used in the calculation of the W and Z self energy in order to calculate $\sin\theta_w$, since this is what is assumed in the two-loop corrections. We use the fine structure constant $\alpha(M_Z)^{\overline{MS}}$, the Z-boson mass M_Z and the Fermi decay constant G_μ as inputs. M_W is predicted from these inputs. Because the EWSB constraints tend to depend sensitively upon $g_{1,2}(M_{QED\times QCD5})$, accurate values for them are determined iteratively. An estimate of the $\overline{DR}$ value of s_W^2 is used to yield a better estimate until the required accuracy is reached (usually within 3 or 4 iterations). The QCD coupling is input as $\alpha_s(M_{QED\times QCD5})^{\overline{MS}}$ and is modified by gluino, squark and top loops as in eqs. (2),(3) of BMPZ in order to obtain the MSSM $\overline{DR}$ value.

$\overline{DR}$ masses are used for all particles in loops in SOFTSUSY, except for M_{A^0} when calculating gauge and Yukawa couplings at $M_{QED\times QCD5}$. At some points in parameter space, the running mass is tachyonic at $M_{QED\times QCD5}$, whereas the pole mass is normal. Using the pole mass at $M_{QED\times QCD5}$ circumvents this problem.

3.4 MSSM Renormalisation

All soft breaking and SUSY parameters are then evolved to the scale M_{SUSY}. By default, we choose the value

$$M_{SUSY} = \sqrt{m_{\tilde{t}_1}(M_{SUSY})m_{\tilde{t}_2}(M_{SUSY})}, \tag{3.9}$$

where [19] the scale dependence of the electroweak breaking conditions is smallest. However, SOFTSUSY allows the user to pick their own value for M_{SUSY}. Throughout the iteration described here, the renormalisation group evolution (RGE) employs three family, 2-loop MSSM β functions for the supersymmetric parameters [2]. $\tan\beta$ and the Higgs VEV parameter v are also run to two-loop order in the Feynman gauge, although the Higgs VEV RGE is missing terms $\mathcal{O}(g_2^4, g_2^2g_1^2, g_1^4)/(16\pi^2)^2$ [20, 21]. There is no step-function decoupling of sparticles: this is taken into account at leading logarithmic order in the radiative corrections previously calculated at M_Z and in the calculation of the physical

[3]It is hoped in the future to include complex phases in Yukawa matrices, sfermion soft mass squared terms and trilinear scalar couplings.

sparticle spectrum at M_{SUSY}, described below. All β functions are real and include 3 family (and mixing) contributions. If no flavour mixing is present in the model specified by the user, the 2-loop parts of the RGEs switch to the dominant 3rd-family version, where the lighter two families' Yukawa couplings are neglected.

3.5 Electroweak Symmetry Breaking

The Higgs VEV parameter $v(M_{QED\times QCD5})$ is set by:

$$v^2(M_{QED\times QCD5}) = 4\frac{M_Z^2 + \Re\Pi^T_{ZZ}(M_{QED\times QCD5})}{g_2^2(M_{QED\times QCD5}) + 3g_1^2(M_{QED\times QCD5})/5}, \tag{3.10}$$

where M_Z is the pole Z mass and Π^T_{ZZ} is the transverse Z self-energy. v is renormalised with two-loop MSSM RGEs to the scale M_{SUSY}, along with the other MSSM parameters. The full one-loop (plus leading two-loop, see below) EWSB conditions at this scale are then employed to[4] calculate $m_3(M_{SUSY})$ and $\mu(M_{SUSY})$. $\mu(M_{SUSY})$ requires an iterative solution because the tadpoles depend upon the value of μ assumed. The symmetry breaking condition for μ can be phrased as [3]

$$\mu^2(M_{SUSY}) = \frac{m^2_{H_1}(M_{SUSY}) - m^2_{H_2}(M_{SUSY})\tan^2\beta(M_{SUSY})}{\tan^2\beta(M_{SUSY}) - 1} - \frac{1}{2}M_Z^2(M_{SUSY}) \tag{3.11}$$

where $m^2_{H_i} = m^2_{H_i} - t_i/v_i$, $M_Z^2(M_{SUSY}) = M_Z^2 + \Re\Pi^T_{ZZ}(M_{SUSY})$ is the running Z mass squared and t_i are the tadpole contributions. The value of μ coming from the tree-level EWSB condition (eq. 3.11, with $\Re\Pi^T_{ZZ} = t_i = 0$) is utilised as an initial guess, then the one-loop contributions in the tadpoles and self-energy terms are added to provide a new value of $\mu(M_{SUSY})$. Two-loop terms $\mathcal{O}(\alpha_t^2)$, $\mathcal{O}(\alpha_b\alpha_\tau)$, $\mathcal{O}(\alpha_b^2)$, $\mathcal{O}(\alpha_b\alpha_s)$, $\mathcal{O}(\alpha_t\alpha_s)$, $\mathcal{O}(\alpha_\tau^2)$ and $\mathcal{O}(\alpha_t\alpha_b)$ are also included in the tadpoles [22, 27]. The tadpole corrections are then calculated using the new value of $\mu(M_{SUSY})$ and the procedure is repeated until it converges to a given accuracy. $m_3(M_{SUSY})$ is then determined by inputting the value of $\mu(M_{SUSY})$ into the EWSB condition

$$m_3^2(M_{SUSY}) = \frac{s_{2\beta}(M_{SUSY})}{2}\left(m^2_{H_1}(M_{SUSY} + m^2_{H_2}(M_{SUSY}) + 2\mu^2(M_{SUSY})\right). \tag{3.12}$$

The ensemble of MSSM parameters are then evolved using the β functions described above to the user supplied scale M_X. If gauge-unification has been specified as a boundary condition, the current estimate of M_X is revised to leading log order to provide a more accurate value upon the next iteration:

$$M_X^{new} = M_X \exp\left(\frac{g_2(M_X) - g_1(M_X)}{g_1'(M_X) - g_2'(M_X)}\right), \tag{3.13}$$

where primes denote derivatives calculated to 2-loop order. The user-supplied boundary conditions are then imposed upon the soft terms before the model is evolved back down to M_{SUSY}. The super particle mass spectrum is determined at this scale. Because μ and m_3 are more scale independent at M_{SUSY} as opposed to some other scale, the Higgs, neutralino and chargino masses also ought to be more scale independent by determining them at this scale.

[4] Note that there is also an option to extract $m_{H_1}(M_{SUSY})$ and $m_{H_2}(M_{SUSY})$ from input $\mu(M_{SUSY})$ and m_A(pole) values, see section E.8.

3.6 MSSM Spectrum

In the following description of the approximations involved in the calculation of the super particle spectrum, it is implicit that where masses appear, their $\overline{DR}$ values are employed. The running value of $s_W(\mu) = e(\mu)/g_2(\mu)$ is also employed. In loop corrections to sparticle masses, the Yukawa couplings of the first two families are set to zero, being highly suppressed compared to those of the third family. All sparticle masses are calculated with the full SUSY one-loop BPMZ corrections at the scale M_{SUSY}. Most sparticle masses are calculated at external momenta equal to their $\overline{DR}$ mass $m(M_{SUSY})$.

The physical gluino mass is calculated to full one-loop order as follows. The running parameters are evaluated at renormalisation scale $\mu = M_{SUSY}$ and external momentum $p = M_3(\mu)$ in the following corrections:

$$\Delta_{\tilde{g}}(\mu) = \frac{g_3(\mu)^2}{16\pi^2} \left(15 + 9\ln\left(\frac{\mu^2}{p^2}\right) - \sum_q \sum_{i=1}^{2} B_1(p, m_q, m_{\tilde{q}_i}, \mu) + \sum_{q=t,b} \frac{m_q}{M_3(\mu)} s_{2\theta_q} \left[B_0(p, m_q, m_{\tilde{q}_1}, \mu) - B_0(p, m_q, m_{\tilde{q}_2}, \mu) \right] \right). \tag{3.14}$$

The Passarino-Veltman functions $B_{0,1}$ are given in appendix B of BMPZ. The physical gluino mass is then given by

$$m_{\tilde{g}} = M_3(M_{SUSY}) \left(1 + \Delta_{\tilde{g}}(M_{SUSY})\right). \tag{3.15}$$

The gluino mass is allowed to be negative, as is the case in mAMSB, for example. Of course the kinematic mass is just the absolute value, and the phase may be rotated away, altering the phases of some of the Feynman rules. Negative masses for neutralinos can also be rotated away in this way.

For the mixed neutralinos and charginos, the external momentum is equal to the $\overline{DR}$ value of the diagonalised mass. All sparticle mixing angles and matrices are defined such that they diagonalise the *tree-level* mass matrix evaluated at M_{SUSY}. When flavour mixing between families is used, at tree-level, the full flavour structure is present in sfermion masses. Loop corrections are added only to the same family entries of the mass squared matrix, not to intra-family mixing entries. For the first two families, the external momentum is set equal to the $\overline{DR}$ sfermion mass itself. For the third family, the external momentum is set equal to the the mass of the lightest $\overline{DR}$ mass eigenvalue of the third family sfermion of the particular flavour in question. All of the `SOFTSUSY` loop corrections also neglect intra-family mixing themselves.

The pseudo-scalar Higgs mass m_{A^0} and CP-even Higgs masses m_{h^0}, m_{H^0} are determined to full one-loop order as in eq. (E.6) of BMPZ in order to reduce their scale dependence, which can be large [23]. The zero-momentum $\mathcal{O}(\alpha_t^2)$, $\mathcal{O}(\alpha_b\alpha_\tau)$, $\mathcal{O}(\alpha_b^2)$, $\mathcal{O}(\alpha_b\alpha_s)$, $\mathcal{O}(\alpha_t\alpha_s)$, $\mathcal{O}(\alpha_\tau^2)$, $\mathcal{O}(\alpha_t\alpha_b)$ 2-loop corrections are also included in $m_{A^0}, m_{h^0}, m_{H^0}$ [24–27]. All one-loop corrections are included in the determination of the charged Higgs pole mass. Every Higgs mass is determined at an external momentum scale equal to its pole mass. α is determined in the limit $p^2 = 0$ (at $p^2 = m_h^2$, the CP-even Higgs mass squared matrix is complex symmetric).

Finally, the running MSSM parameters are evolved back down to $M_{QED\times QCD5}$. The whole process is iterated as shown in figure 3.2, until the $\overline{DR}$ sparticle masses evaluated at M_{SUSY} all converge to better than the desired fractional accuracy (`TOLERANCE`), which may be set by the user in the main program or input file. The predicted value of M_Z is required to be in the domain $(1 \pm 10\times$`TOLERANCE`$)M_Z$.

3.7 Fine Tuning

We now detail the fine-tuning calculation. As lower bounds on super partner masses are pushed up by colliders, m_{H_1} and m_{H_2} may be forced to be much larger than M_Z if they are related to the other super particle masses, as is the case for example in the case of minimal super gravity. If we re-phrase eq. (3.11) as

$$M_Z^2 = -2\mu^2 + \tan 2\beta \left[m_{H_2}^2 \tan\beta - m_{H_1}^2 \cot\beta \right], \tag{3.16}$$

we see that the terms on the right-hand side must have some degree of cancellation in order to reproduce the observed value of M_Z. But μ has a different origin to the SUSY breaking parameters and the balancing appears unnatural. Various measures have been proposed in order to quantify the apparent cancellation, for example refs. [28, 29]. The definition of naturalness c_a of a 'fundamental' parameter a employed here is [29]

$$c_a \equiv \left| \frac{\partial \ln M_Z^2}{\partial \ln a} \right|. \tag{3.17}$$

From a choice of a set of fundamental parameters defined at the scale M_X: $\{a_i\}$, the fine-tuning of a particular model is defined to be $c = \max(c_a)$. $\{a_i\}$ are any parameters in the user supplied boundary condition on the soft supersymmetry breaking parameters augmented by $h_t(M_X), \mu(M_X)$ and $m_3(M_X)$. The derivatives in eq. (3.17) are calculated by numerically finding the derivative of $M_Z^{pole} = \hat{M}_Z + \Re\Pi_{ZZ}^T(M^2_{\tilde{Q}ED\times QCD5})$ in eq. (3.11). The input parameters are changed slightly (one by one), then the MSSM parameter ensemble is run from M_X to M_{SUSY} where the sparticle mass spectrum is determined along with the corresponding $\overline{MS}$ Higgs VEV parameter $v^2 \equiv v_1^2 + v_2^2$. First of all, $\tan\beta(M_{SUSY})$ is determined by inverting eq. (3.12) and the resulting value is utilised in a version of eq. (3.11) inverted to give M_Z^{pole} in terms of the other parameters. The resulting value of M_Z^{pole} is the prediction for the new changed input parameters, and its derivative is determined by examining its behaviour as the initial changes in input parameters tend to zero.

A Running SOFTSUSY

A main program which produces an executable called `softpoint.x`, is included in the `SOFTSUSY` distribution. For the calculation of the spectrum of single points in parameter space, we recommend the SLHA [30] input/output option. The user must modify a file (e.g. `lesHouchesInput`, as provided in the standard distribution) that specifies the input parameters. The user may then run the code with

```
./softpoint.x leshouches < lesHouchesInput
```

The output will also be in SLHA format. Such output can be used for input into other programs which subscribe to the accord, such as `SDECAY` [37], `PYTHIA` [38] (for simulating sparticle production and decays at colliders) or `micrOMEGAs` [39] (for calculating the relic density of neutralinos, $b \to s\gamma$ and $\mu \to e\gamma$), for example. For further details on the necessary format of the input file, see ref. [30] and appendix A.1.

As of `SOFTSUSY 3.4`, the command line interface of `softpoint.x` has changed of the user does not wish to use the SLHA, see `softpoint.x --help`. For the MSSM the new syntax is

```
./softpoint.x <susy-breaking-model> [susy-breaking-model options] [general options]
```

where the available SUSY-breaking models are listed in Table 2 and the general options are listed in Table 3. Bracketed entries should be replaced by the desired numerical values, (in GeV if they are

susy-breaking model	susy-breaking model option	description
`sugra`	`--m0=<value>`	unified soft scalar mass
	`--m12=<value>`	unified soft gaugino mass
	`--a0=<value>`	unified trilinear coupling
`amsb`	`--m0=<value>`	unified soft scalar mass
	`--m32=<value>`	gravitino mass
`gmsb`	`--n5=<value>`	number of $SU(5)$ mulitplets
	`--mMess=<value>`	messenger scale
	`--LAMBDA=<value>`	SUSY breaking scale $< F_S > / < S >$
	`--cgrav=<value>`	Gravitino mass scaling factor

Table 2. MSSM command line options for `softpoint.x`. The SUSY breaking model parameters should instead be fixed, because often they have zero default values, and failing to set them could result in an unphysical point.

general option	description
`--tol=<value>`	numerical convergence precision
`--mbmb=<value>`	bottom mass $m_b^{\overline{MS}}(m_b)$
`--mt=<value>`	top mass $m_t^{\rm pole}$
`--alpha_s=<value>`	strong coupling $\alpha_s^{\overline{MS}}(M_Z)$
`--QEWSB=<value>`	EWSB scale
`--alpha_inverse=<value>`	inverse electromagnetic cougling $1/\alpha_{\rm em}^{\overline{MS}}(M_Z)$
`--tanBeta=<value>`	$\tan\beta$
`--sgnMu=<value>`	$\mathrm{sgn}(\mu)$
`--mgut=unified`	define unification scale M_X by $g_1(M_X) = g_2(M_X)$
`--mgut=msusy`	set unification scale to $M_{\rm SUSY}$
`--mgut=<value>`	set unification scale to fixed value
`--higgsUncertainties`	provides an estimate of uncertainties in Higgs masses
`--matchingScale=<value>`	set $M_{QED\times QCD5}$

Table 3. General command line options for `softpoint.x` These variables all have default values if they are omitted.

dimensionful). The program will provide output from one point in the CMSSM, mAMSB or mGMSB model.

For users that are not familiar with `C++`, we note that the executable interface allows the calculation at just one parameter point in SUSY breaking space. If scans are required, the user can either call `SOFTSUSY` from a shell script or use a system call from a main `C` program to the executable. Alternatively, a main program showing an example of a scan is provided. `C++` beginners should note in the following that "method" means function, that objects contain a list of data structures and functions and that for a user to access (change or reference) the data encoded in an object, one of its functions should be called. Such functions are given in tables of the following appendices.

A.1 Input file

If, as recommended, the SLHA option is used for input, the user may add a `SOFTSUSY`-specific block to the input file in the following format, with bracketed entries replaced by double precision values:

```
Block SOFTSUSY     # SOFTSUSY specific inputs
  1   <TOLERANCE>  # desired fractional accuracy in output
  2   <mixing>     # quark mixing option
  3   <PRINTOUT>   # gives additional verbose output during calculation
  4   <QEWSB>      # change electroweak symmetry breaking scale
  6   <PRECISION>  # number of significant figures in SLHA output
  7   <numHiggsLoops> # number of loops in REWSB/mh calculation
 10   <forceSlha1>    # if =1, tries to force output into SLHA 1 format
 11   <m32>           # set gravitino mass
 12   <printSpectra>  # Prints spectrum even when point is theoretically
excluded if=1
 13   <mAflag>        # If =0 (default), sets tachyonic mA=0, otherwise mA=sqrt(|mA|^2)
 21   <higgsUncert>   # If =1, provides an estimate of Higgs mass uncertainties
 27   <mQEDxQCD5>      # Scale at which QEDxQCD is matched to MSSM (default mt)
 28   <mh0(pole)>      # Artificially fixed value of mh0(pole)
 29   <mA0(pole)>      # Artificially fixed value of mA0(pole): TRUMPS EXTPAR 26
 30   <mH0(pole)>      # Artificially fixed value of mH0(pole)
 31   <mH+/-(pole)>    # Artificially fixed value of mH+/-(pole)
```

The fractional numerical precision on masses and couplings output by SOFTSUSY is better than TOLERANCE, which sets the accuracy of the whole calculation. The iteration of each physical SUSY particle mass is required to converge to a fractional accuracy smaller than TOLERANCE. Sub-iterations are required to converge to a better accuracy than $10^{-2}\times$TOLERANCE for s_W and $10^{-4}\times$TOLERANCE for μ. The accuracy of the Runge-Kutta RGE changes from iteration to iteration but is proportional to the value of TOLERANCE. Values between 10^{-2} and 10^{-6} are common, lower values mean that SOFTSUSY takes significantly longer to perform the calculation.

The next parameter mixing determines what M_Z boundary condition will be used for the quark Yukawa matrix parameters. mixing=0.0 sets the quark mixings to zero but includes the first two family's diagonal terms. mixing=1.0,2.0 sets all the mixing at M_Z to be in the up-quark or down-quark sector respectively, as in eq. (3.8). mixing=-1.0 sets the first two family Yukawa couplings to be zero. This last option should *not* be used if flavour mixing is desired, and is only included for backwards compatibility/checks.

Setting PRINTOUT to a non-zero value gives additional information on each successive iteration. If PRINTOUT>0, a warning flag is produced when the overall iteration finishes. The predicted values of M_Z^{pole} and $\tan\beta(M_{SUSY})$ after iteration convergence are also output[5]. The level of convergence, $\mu(M_{SUSY})$, $m_3^2(M_{SUSY})$ and M_Z are output with each iteration, as well as a flag if the object becomes non-perturbative. PRINTOUT>1 produces output on the fine-tuning calculation. The predicted values of M_Z^{pole} and $\tan\beta(M_{SUSY})$ are output with each variation in the initial inputs. A warning flag is produced when a negative-mass squared scalar is present. PRINTOUT>2 prints output on the sub-iterations that determine $\mu(M_{SUSY})$ and $s_W(M_{SUSY})$, and flags the nature of any tachyons encountered. Values PRINTOUT>0 are only required if additional diagnostics are required for debugging purposes.

Switching on the 2-loop corrections (1) results in a full 2-loop RGE evolution, but slows the calculation by a factor of approximately three. <numHiggsLoops> may be set to either 2 (default) or 1, and is the number of loops of particles allowed to contribute to the Higgs mass threshold calculation and the tadpoles in the electroweak symmetry breaking conditions. <mAflag> determines SOFTSUSY's behaviour when it calculates that the tree-level value of $m_{A^0}^2(\mu) < 0$ for $\mu = M_Z$ or $\mu = M_{SUSY}$, where

[5]Note that the input value of $\tan\beta$ is the value at M_Z.

various loop corrections involving the A^0 boson are needed. For the default value `0`, $m_{A^0}(\mu) = 0$ is set (and the tachyonic A^0 flag is raised). For the other possible value `1`, $m_{A^0}(\mu)$ is set to be the square root of the absolute value of $m^2_{A^0}(\mu)$, as far as the loop correction is concerned, and the tachyonic A^0 flag is raised. It is not entirely clear (to the author at least), whether tachyonic tree-level $m^2_{A^0}$ values are an indication of an unphysical parameter point or not. Regardless, some points in parameter space report tachyons only in the early iterations of the algorithm depicted in Fig. 3.2, and such points may be physically fine if there are no further problems reported.

`<higgsUncert>` is by default set to `0`, but if it is set to `1`, then `SOFTSUSY` will estimate the approximate errors in each Higgs boson mass, in units of GeV. This is printed out at the bottom of the SLHA file in block called `DMASS`. Sources of error in Higgs mass predictions are added *linearly*. They are calculated from individually varying M_{SUSY}, $M_{QED\times QCD5}$ by a factor or 2 in each direction and taking the maximum and minimum Higgs mass values within the range. Other finite uncertainties are taken into account by changing the determination of $m_t(Q)$, $\alpha(Q)$ and $\alpha_s(Q)$ by higher order effects [40].

`<mQEDxQCD5>` is equal to the scale at which QED×QCD are matched to the MSSM $M_{QED\times QCD5}$, which by default is m_t.

`<mh0(pole)>`, `<mA0(pole)>`, `<mA0(pole)>` and `<mH+/-(pole)>` are intended for users that wish to use another speciality program to calculate pole masses of the Higgs bosons. They may use the other program to calculate them and then input them within these variables, in units of GeV. `SOFTSUSY` will perform its spectrum calculation as usual, but will overwrite the pole Higgs boson masses at the end of the calculation with the relevant inputted values. Decays will be calculated using these input masses.

In the standard SLHA input format, a parameter `12` in `Block MODSEL` fixes the maximum scale of printing out the running parameters, Q_{max}. `SOFTSUSY` has an additional option to the usual SLHA definition: if the parameter is set to `-1.000000`, then the GUT scale is taken for Q_{max}.

If `softpoint.x` is used *without* the SLHA interface, default Standard Model inputs are used from the files `def.h` and `lowe.h`. The low energy data is encoded in a `QedQcd` object and must be provided. The default numbers supplied and contained in the `QedQcd` object are given in units of GeV and running masses are in the $\overline{MS}$ scheme. For the bottom and top masses, *either* the running mass *or* the pole mass must be supplied as an input. The type of mass not given for input is calculated by `SOFTSUSY` at the 3-loop QCD level. We recommend, along with ref. [33], that the running mass be used for m_b since there are smaller theoretical errors in the extraction of this quantity from experiment than the pole mass. The scale dependent quantities in this object are then evolved to M_Z by the method `toMz`, to provide the low-scale empirical boundary condition for the rest of the calculation. The MSSM spectrum calculated depends most crucially upon $M_Z, \alpha(M_Z), \alpha_s(M_Z)$ and input third family fermion masses.

B Sample Program

We now present the sample program from which it is possible to run `SOFTSUSY` in a simple fashion. The program we presents here performs a scan in the variable $\tan\beta$, with other parameters as in point CMSSM10.1.1 [34]. It then prints the four pole Higgs masses as a function of $\tan\beta$ in the standard output channel along with a Bayesian measure of naturalness [35]. If there are any problems with the parameter point, the program prints out these instead of the Higgs masses. The most important features of the objects are described in appendix E. The sample program `main.cpp` has the following form:

```
/**
   Project:     SOFTSUSY
   File:        main.cpp
   Author:      Ben Allanach
   Manual:      B.C. Allanach, hep-ph/0104145, Comp. Phys. Comm. 143 (2002) 305
                B.C. Allanach, M.A. Bernhardt, arXiv:0903.1805, Comp. Phys.
                Commun. 181 (2010) 232-245
   Webpage:     http://allanach.home.cern.ch/allanach/softsusy.html
   Description: main calling program example: performs a scan of tan beta
   (starting at CMSSM10.1.1) and prints out Higgs masses as a result in
   the format
       <tan beta>      <mh0>     <mA0>    <mH0>     <mH+/->
*/

#include "main.h"
#include "nmssmsoftsusy.h"
#include "decays.h"

#ifdef HAVE_CONFIG_H
#include "config.h"
#endif

using namespace softsusy;

int main() {
  /// Sets up exception handling
  signal(SIGFPE, FPE_ExceptionHandler);

  try {
    /// Sets format of output: 6 decimal places
    outputCharacteristics(6);

    cout << "# SOFTSUSY" << PACKAGE_VERSION
         << " test program, Ben Allanach\n";
    cout << "# If you use SOFTSUSY, please refer to B.C. Allanach,\n";
    cout << "# Comput. Phys. Commun. 143 (2002) 305, hep-ph/0104145\n";

    /// Parameters used: CMSSM parameters
    double m12 = 500., a0 = 0., mGutGuess = 2.0e16, tanb = 10.0, m0 = 125.;
    int sgnMu = 1;       ///< sign of mu parameter
    int numPoints = 10;  ///< number of scan points

    QedQcd oneset;       ///< See "lowe.h" for default definitions parameters

    /// most important Standard Model inputs: you may change these and recompile
    double alphasMZ = 0.1187, mtop = 173.5, mbmb = 4.18;
    oneset.setAlphaMz(ALPHAS, alphasMZ);
    oneset.setPoleMt(mtop);
    oneset.setMbMb(mbmb);
```

```
    oneset.toMz();       ///< Runs SM fermion masses to MZ
    oneset.runto(oneset.displayPoleMt()); ///< Run them to mt for matching

    /// Print out the SM data being used, as well as quark mixing assumption and
    /// the numerical accuracy of the solution
    cout << "# Low energy data in SOFTSUSY: mixing=0" << " TOLERANCE="
         << TOLERANCE << endl;

    /// Print out header line
    cout << "# tan beta   mh           mA           mH0          mH+-         BN\n";

    int i;
    /// Set limits of tan beta scan
    double startTb = 2., endTb = 50.;

    /// Cycle through different points in the scan
    for (i = 0; i <= numPoints; i++) {
      tanb = (endTb - startTb) / static_cast<double>(numPoints) *
           static_cast<double>(i) + startTb;  /// set tan beta ready for the scan.

    /// Preparation for calculation: set up object and input parameters
    MssmSoftsusy r;

    DoubleVector pars(3);
    pars(1) = m0; pars(2) = m12; pars(3) = a0;
    bool uni = true;  ///< MGUT defined by g1(MGUT)=g2(MGUT)

    /// Calculate the spectrum
    r.lowOrg(sugraBcs, mGutGuess, pars, sgnMu, tanb, oneset, uni);

    /// check the point in question is problem free: if so print the output
    if (!r.displayProblem().test())
      cout << tanb << " " << r.displayPhys().mh0(1) << " "
           << r.displayPhys().mA0(1) << " "
           << r.displayPhys().mh0(2) << " "
           << r.displayPhys().mHpm << " "
           << r.calcBayesianNaturalness() << endl;
    else
      /// print out what the problem(s) is(are)
      cout << tanb << " " << r.displayProblem() << endl;
  }
  }
  catch(const string & a) { cerr << a; return -1; }
  catch(const char * a) { cerr << a; return -1; }
  catch(...) { cerr << "Unknown type of exception caught.\n"; return -1; }

  return 0;
}
```

First of all, a function `signal` is called which attempts to catch any floating point exceptions that may

name	arguments
`sugraBcs`	m_0, $m_{1/2}$, A_0
`amsbBcs`	$m_{3/2}$, m_0
`gmsbBcs`	n_5, m_{mess}, Λ

Table 4. SUSY breaking boundary conditions available to the user, detailing arguments in order. The asterisk denotes additional information in the text.

occur during the running of the program. Then, after an initial introductory print-out, the variables specifying the supersymmetry breaking parameters are specified. For these, the same notation as appendix A is used. Next, the important Standard Model inputs are defined and combined with the defaults already present in the `QedQcd` object. The top running mass is calculated from the pole mass and Standard Model fermion masses and gauge couplings are then run up to M_Z with the method `toMz`. They are then run to the pole top mass with the method `runto`.

If `gaugeUnification=true`, `softsusy` will determine `mGutGuess` from electroweak gauge unification, using the `mGutGuess` value supplied as an initial guess. The user can supply a void function that sets the supersymmetry breaking parameters from an input `DoubleVector`. In the sample code given above, this function is `sugraBcs` and is applied to the `MssmSoftsusy` object at the scale M_{GUT}, which will be determined by the user (but `mGutGuess` will be used as an initial guess). Other examples of available boundary conditions are given in Table 4. The user must supply a `DoubleVector` containing the numerical values of the arguments, correctly ordered as in Table 4. `sugraBcs`, for example, calls the `MssmSoftsusy` method `standardSugra(m0, m12, a0)`, which sets all scalar masses equal to `m0`, all gaugino masses to[6] `-m12` and all trilinear scalar couplings to `a0`, in the standard universal fashion. If the user desires to write his or her own boundary condition, it must conform to the prototype

```
void userDefinedBcs(MssmSoftsusy & m, const DoubleVector & inputs)
```

The method `lowOrg` drives the calculation, after which $\tan\beta$ and the Higgs masses are printed out.

One other small extension to the SLHA input is the option of inputting negative SUSY breaking scalar mass squared values as the theoretical boundary condition at the input scale of the theoretical boundary condition. If this is desired, one writes the *negative* of the mass in the SLHA input file.

C Sample Output

For the recommended SLHA option, the conventions for the output are explained in Ref. [30]. We present the non-SLHA compliant `SOFTSUSY` output for the main program above, which can be run by the command

```
./softsusy.x
```

The output obtained was

```
# SOFTSUSY4.1.24 test program, Ben Allanach
# If you use SOFTSUSY, please refer to B.C. Allanach,
# Comput. Phys. Commun. 143 (2002) 305, hep-ph/0104145
# Low energy data in SOFTSUSY: mixing=0 TOLERANCE=1.000000e-04
# tan beta   mh           mA           mH0          mH+-         BN
2.000000e+00 9.445510e+01 1.024076e+03 1.026621e+03 1.027472e+03 1.419678e+02
```

[6] With our sign conventions, the CMSSM should impose a common negative gaugino mass.

```
6.800000e+00 1.135516e+02 7.378767e+02 7.384303e+02 7.425306e+02 9.997849e+01
1.160000e+01 1.152140e+02 7.128471e+02 7.130515e+02 7.175897e+02 7.312781e+01
1.640000e+01 1.156133e+02 6.938823e+02 6.939918e+02 6.987302e+02 3.594097e+01
2.120000e+01 1.157416e+02 6.726202e+02 6.726915e+02 6.776181e+02 2.761242e+01
2.600000e+01 1.157684e+02 6.477811e+02 6.478349e+02 6.529844e+02 1.294738e+02
3.080000e+01 1.157262e+02 6.191206e+02 6.191670e+02 6.245949e+02 2.866695e+02
3.560000e+01 1.155877e+02 5.866149e+02 5.866576e+02 5.924371e+02 5.247920e+02
4.040000e+01 1.151904e+02 5.502525e+02 5.502822e+02 5.565225e+02 1.364838e+03
4.520000e+01 [ stau tachyon ]
5.000000e+01 [ stau tachyon ]
```

Firstly, the output details the input parameters, starting with `mixing` and `TOLERANCE`. After the output of the input `QedQcd` object, various pole Higgs masses are displayed for different values of tan beta, as labelled by the various columns. The columns are, respectively, $\tan\beta$, m_{h^0}, m_{A^0}, m_{H^0}, $m_{H^\pm}$ and a measure of Bayesian naturalness [35]. The returned `nan` value indicates that the naturalness calculation had numerical instabilities at the point in question. In the case that there is a different problem with the point, the higgs masses are replaced by a warning: three points have tachyonic staus, and the last has not achieved convergence of the iteration and possesses a negative mass squared for $m_{H^\pm}$. Here, `stau tachyon` indicates that the stau mass has become imaginary and so the scalar potential minimum does not conserve electromagnetism. For the highest value of $\tan\beta$, `No convergence` indicates that `SOFTSUSY` was not able to iterate the calculation to the desired accuracy. `hpm tachyon` indicates that the charged Higgs masses have become imaginary and the scalar potential minimum does not conserve electromagnetism. The presence of such tachyons rules such parameter points out. We list and explain all of the possible problem flags in the following section.

C.1 Problem flags

Any associated problems such as negative mass-squared scalars or inconsistent EWSB are flagged at the end of the output. We now list the problems, indicating their meaning:

- If `No convergence` appears, then `SOFTSUSY` is indicating that it didn't achieve the accuracy of `TOLERANCE` within less than 40 iterations. The output of the code is therefore to be considered unreliable and it is not clear from the output whether the point is allowed or disallowed, despite the presence or absence of other warning messages. This error flag often appears near the boundary of electroweak symmetry breaking, (where $\mu(M_{SUSY}) = 0$)), where the iterative algorithm is not stable. To calculate the position of the electroweak symmetry boundary, one should interpolate between regions a small distance away from it.

- `Non-perturbative` indicates that `SOFTSUSY` encountered couplings reaching Landau poles when evolving, and could not calculate any further. Any results obtained using perturbation theory (for example those of `SOFTSUSY`) therefore cannot be trusted.

- `Infra-red quasi fixed point breached` indicates that the parameter point is at a Landau pole of a Yukawa coupling. This should not be a problem provided no other errors are flagged.

- `muSqWrongSign` indicates that the Higgs minimisation conditions imply that $\mu^2 < 0$, meaning that the desired electroweak minimum is not present in the model. The model is ruled out.

- `m3sq` indicates that m_3^2 from eq. (3.12) has the incorrect sign, meaning that the desired electroweak minimum is not present in the model. The model is ruled out.

- The `tachyon` variable labels if a scalar particle other than the Higgs has acquired a negative mass squared, when $M_Z^2 < 0$ or when a pole Higgs masses is imaginary. The model is ruled out. `tachyonWarning`, on the other hand, denotes that a tree-level scalar mass squared has run to be negative at the renormalisation scale equal to M_Z. Since these tree-level masses are used in loops to calculate various other quantities (including the $Z-$boson mass) and fix gauge and Yukawa couplings, it is possible that the numerical predictions are somewhat inaccurate (`SOFTSUSY`will use almost zero masses for these tachyons). So: the model is not necessarily ruled out, and the flag results in a warning. are used in loops to calculate `tachyon` is of an enumerated type `tachyonType`, which relabels the integers $z \in (0,\ 17)$ to indicate which particle is a tachyon according to the rule

  $$0 = \text{no tachyon},\ 1 = \tilde{e},\ 2 = \tilde{\mu},\ 3 = \tilde{\tau},\ 4 = \tilde{u},\ 5 = \tilde{c},\ 6 = \tilde{c},\ 7 = \tilde{d},\ 8 = \tilde{s},$$
  $$9 = \tilde{b},\ 10 = h^0,\ 11 = A^0,\ 12 = H^{\pm},\ 13 = \tilde{\nu}_e,\ 14 = \tilde{\nu}_\mu,\ 15 = \tilde{\nu}_\tau,\ 16 = W^{\pm},\ 17 = Z^0$$

  respectively.

- `noRhoConvergence` is flagged when `SOFTSUSY` cannot calculate the ρ parameter and determine gauge couplings from data, typically because of infinities that have crept into the calculation. The other problems are serious enough to rule the model out.

- `higgsUfb` and `bProblem` indicate that the desired electroweak minimum is in fact a saddle point of the potential, thus the model is ruled out.

- `mgutOutOfBounds` is flagged if the value of the gauge unification scale predicted by eq. 3.13 is outside the range 10^4 GeV $< M_X < 5\times10^{17}$ GeV. The GUT-scale has been set to the appropriate limit, and the `SOFTSUSY` numbers cannot be trusted.

- `inaccurateHiggsMass` is flagged when the $\overline{DR}$ perturbation series has broken down, and higher order terms are potentially $\sim \mathcal{O}(1)$. This can happen when $(M_3/m_{\tilde{t}_1})^2 > 16\pi^2$, or when $(\mu/m_{\tilde{t}_1})^2 > 16\pi^2$ [24]. The rest of the calculation should be fine, but one cannot trust the higgs masses. One should use some other program that uses the on-shell scheme to calculate Higgs masses for these points[7].

Thus flags other than `No convergence`, `Infra-red quasi fixed point breached`, inaccurateHiggsMass or `Non-perturbative` indicate an unphysical minimum of the scalar potential, effectively ruling the model point out. Another structure within `MssmSoftsusy` of type `sProblem` flags various potential problems with the object, for example the lack of radiative EWSB or negative mass squared scalars (excluding the Higgs mass squared parameters). This structure is shown in table 13. In addition, the method `test` prints out if any of the possible data variables flagging problems are true. The `higgsUfb` flag is true if

$$m_{H_1}^2 + 2\mu^2 + m_{H_2}^2 - 2|m_3^2| < 0 \tag{C.1}$$

is not satisfied, implying that the desired electroweak minimum is either a maximum or a saddle-point of the tree-level Higgs potential [2]. The contents of `sPhysical` and `sProblem` can be output with overloaded `<<` operators. `noConvergence` means that the desired accuracy was not reached. `nonPerturbative` or `irqfp` flags the existence of a Landau pole in the renormalisation group evolution, and the calculation is not perturbatively reliable so any results should be discarded. All other

[7] Although we note that there are many different points for which the $\overline{OS}$ scheme has a perturbation series problem and one must use the $\overline{DR}$ instead.

variable	default	description
ARRAY_BOUNDS_CHECKING	off	Vector and Matrix bounds checking
EPSTOL	10^{-11}	Underflow accuracy
GMU	$1.16637\ 10^{-5}$	G_μ, Fermi constant from muon decay
MZCENT	91.1876	Pole mass of the Z^0 boson M_Z.

Table 5. Switches and constants. Starred entries have more explanation in the text. G_μ is in units of GeV^2 and M_Z in GeV.

problems except noConvergence and nonPerturbative should be considered as grounds for ruling the model out. noRhoConvergence occurs when the pseudo-scalar Higgs A^0 has a negative mass squared (i.e. an invalid electroweak vacuum).

D Switches and Constants

The file def.h contains the switches and constants. If they are changed, the code must be recompiled in order to use the new values. def.cpp contains initial values for global variables (in the SOFTSUSY namespace). These may be changed by the user in their main programs. Table 5 shows the most important parameters in def.h, detailing the default values that the constants have. G_μ and M_Z have been obtained using the latest particle data group numbers [17].

E Object Structure

We now go on to sketch the objects and their relationship. This is necessary information for generalisation beyond the MSSM. Only methods and data which are deemed important for prospective users are mentioned here, but there are many others within the code itself.

E.1 Linear Algebra

The SOFTSUSY program comes with its own linear algebra classes: Complex, DoubleVector, DoubleMatrix, ComplexVector, ComplexMatrix. Constructors of the latter four objects involve the dimensions of the object, which start at 1. Complex objects are constructed with their real and imaginary parts respectively. For example, to define a vector $a_{i=1,2,3}$, a matrix $m_{i=1\ldots3,j=1\ldots4}$ of type double and a Complex number $b = 1 - i$:

```
DoubleVector a(3);
DoubleMatrix m(3, 4);
Complex      b(1.0, -1.0);
```

Obvious algebraic operators between these classes (such as multiplication, addition, subtraction) are defined with overloaded operators *, +, - respectively. Elements of the vector and matrix classes are referred to with brackets (). DoubleVector and DoubleMatrix classes are contained within each of the higher level objects that we now describe.

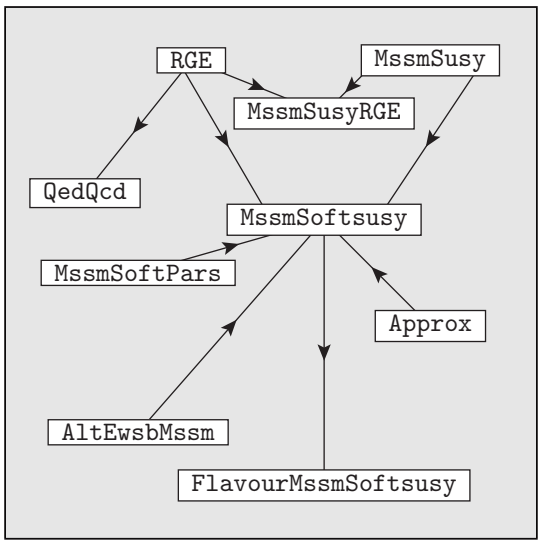


Figure 2. Heuristic high-level object structure of `SOFTSUSY`. Inheritance is displayed by the arrows and a type definition by the equals sign.

E.2 General Structure

From a RGE point of view, a particular quantum field theory model consists of a set of couplings and masses defined at some renormalisation scale μ. A set of β functions describes the evolution of the parameters and masses to a different scale μ'. This concept is embodied in an *abstract* `RGE` object, which contains the methods required to run objects of derived classes to different renormalisation scales. The other objects displayed in figure 2 are particular instances of `RGE`, and therefore inherit from it. `QedQcd` objects consist of data on the quark and lepton masses and gauge couplings. They contain the β functions for running in an effective QED×QCD theory below m_t. An object of class `MssmSusy` contains the Yukawa couplings, and the three gauge couplings of the MSSM. It also contains the superpotential μ term (not to be confused with the renormalisation scale), $\tan\beta$, the ratio of the two Higgs doublet VEVs as well as $v = \sqrt{v_1^2 + v_2^2}$. Its β functions are valid in the exact SUSY limit of the MSSM. The major part of the code resides within the `MssmSoftsusy` class. Objects of this type have all the functionality of `MssmSusy`, with soft SUSY breaking terms and theoretical boundary conditions contained in the inherited class `SoftParsMssm`. It also contains an object of type `QedQcd` which contains weak scale empirical data. Code in the `MssmSoftsusy` class organises and performs the main part of the calculation. `FlavourMssmSoftsusy` objects are also slight variants: they have full flavour-mixed output and input. In the following, we provide basic information on classes so that users may program using `SOFTSUSY`. Highly detailed and technical documentation on the program may be obtained from the `SOFTSUSY` website.

E.3 `RGE` Class

The data and important methods in `RGE` are presented in table 6. Each of the higher level objects described in this appendix have explicitly named `display` and `set` methods that are used to access or change the data contained within each object. In table 6 (as in the following tables in this section), these accessing methods are listed on the same row as the relevant data variable.

data variable		methods
`double mu`= μ	renormalisation scale (GeV)	`setMu` `displayMu`
`int numpars`	number of scale dependent parameters	`setPars` `howMany`
method	function	
`DoubleVector display()`	displays all running parameters (*)	
`void set(DoubleVector)`	sets all running parameters (*)	
`DoubleVector beta`	displays beta functions of all running parameters (*)	
`runto`	runs object to new value of `mu`	

Table 6. Abstract `RGE` class. (*) indicates that derived objects *must* contain these methods (see text).

data variable		methods
`int loops`	accuracy of RGE	`setLoops` `displayLoops`
`int thresholds`	accuracy level of threshold computation	`setThresholds` `displayThresholds`

Table 7. `Approx` class.

The `RGE` method `runto(mup, eps)` will automatically run any derived object to the scale `mup` with a fractional accuracy of evolution `eps`. In order to define this evolution, any object that inherits from an `RGE` must contain three methods: `display`, `set`, `beta` shown in table 6. `DoubleVector display() const` must return a vector containing all masses and couplings of the object, in some arbitrary user-defined order. `void set(const DoubleVector & v)` must set these couplings given a `DoubleVector v` defined in the same order as the display function. `DoubleVector beta() const` must then return the β functions in a `DoubleVector` defined as

$$\beta_i = \frac{da_i}{d\ln\mu}, \tag{E.1}$$

where a_i denotes any mass or coupling of the model. The ordering of the a_i must be identical in each of the three methods.

E.4 `Approx` Class

`Approx` objects are for groups of couplings and masses that have renormalisation group equations and have a number of loops `loops` that RGEs are calculated at, and an integer `thresholds` specifying which terms to include in any threshold computations (usually the higher the integer the more corrections are included). We show the class' relevant parameters and methods in Table 7.

E.5 `QedQcd` Class

The `QedQcd` class contains a `DoubleVector` of quark and lepton $\overline{MS}$ masses ($m_f = m_{u,d,e,c,s,\mu,t,b,\tau}(\mu)$), as shown in table 8. Its contents may be printed to standard output or read from standard input (with the same format in each case) by using the operators `<<` or `>>`, as can all the non-abstract objects mentioned in this section. The methods `toMz()`, `toMt()` act on an initial object defined with each

data variable		methods
`DoubleVector a`	$\overline{MS}$ gauge couplings	`setAlpha`
$\alpha(\mu), \alpha_s(\mu)$		`displayAlpha`
`DoubleVector m`	running fermion masses	`setMass`
$m_f(\mu)$	vector $(1 \ldots 9)$ (GeV)	`displayMass`
`double mtPole, mbPole`	pole top/bottom/tau	`setPoleMt`, `setPoleMb`
`double mtauPole`	mass	`setPoleMtau`
$m_t^{pole}, m_b^{pole}, m_\tau^{pole}$	(GeV)	`displayPoleMt displayPoleMb`
		`displayPoleMtau`

method	function
`runGauge`	runs gauge couplings *only*
`toMt, toMZ`	runs fermion masses and gauge couplings from Q' to m_t^{pole} or M_Z

Table 8. `QedQcd` class. Q' is defined in the text.

fermion mass m_f defined at a scale

$$Q' = \max(1 \text{ GeV}, m_f(m_f)) \tag{E.2}$$

and gauge couplings at M_Z.

E.6 `MssmSusy` Class

The operators `<<`, `>>` have been overloaded to write or read a `MssmSusy` object to/from a file stream. Table 9 shows the data variables and important methods contained in the class. For the Yukawa and gauge couplings, methods exist to either set (or display) one element or a whole matrix or vector of them. Another object, `MssmSusyRGE` is also available, which is an `RGE` object and therefore can be run to different scales. `mixing` determines what M_Z boundary condition will be used for the quark Yukawa matrix parameters. `mixing`=0.0 sets the quark mixings to zero but includes the first two family's diagonal terms. `mixing`=1.0,2.0 sets all the mixing at M_Z to be in the up-quark or down-quark sector respectively, as in eq. (3.8). `mixing`=-1.0 sets the first two family Yukawa couplings to be zero. This last option should *not* be used if flavour mixing is desired, and is only included for backwards compatibility/checks.

E.7 `SoftParsMssm` Class

The operators `<<`, `>>` have been overloaded to write or read a `softParsMssm` object to/from a file stream. Table 10 shows the data variables and important methods contained in the class. `addAmsb()` adds anomaly mediated supersymmetry breaking terms [41] to the model's soft parameters. Such terms are proportional to the VEV of a compensator superfield, so $m_{3/2}$ in table 10 must have been set before `addAmsb` is used. `minimalGmsb(int n5, double LAMBDA, double mMess)` applies the messenger scale `mMess` boundary conditions to the soft masses in minimal gauge-mediated supersymmetry breaking [42]. `n5` denotes the number of $5 \oplus \bar{5}$ messenger fields that are present and `LAMBDA`(Λ) is as described in ref. [42].

data variable		methods
`int mixing`	quark mixing parameter (default 0)	`setMixing` `displayMixing`
`Approx mssmSusyApprox`	loops/thresholds	`setMssmApprox` `displayMssmApprox`
`DoubleMatrix u, d, e` $(Y_U)_{ij}, (Y_D)_{ij}, (Y_E)_{ij}$	Yukawa couplings (3 by 3 matrix)	`setYukawaElement` `setYukawaMatrix` `displayYukawaElement` `displayYukawaMatrix`
`DoubleVector g` g_i	MSSM gauge couplings $(1 \ldots 3)$ vector	`setAllGauge` `setGaugeCoupling` `displayGauge` `displayGaugeCoupling`
`double smu` μ	bilinear Higgs superpotential parameter	`setSusyMu` `displaySusyMu`
`double tanb` $\tan\beta$	ratio of Higgs VEVs (at current renormalisation scale)	`setTanb` `displayTanb`
`double hVev` v	Higgs VEV	`setHvev` `displayHvev`

method	function
`setDiagYukawas`	calculates and sets all diagonal Yukawa couplings given fermion masses and a Higgs VEV
`getMasses`	calculates quark and lepton masses from Yukawa couplings
`getQuarkMixing`	mixes quark Yukawa couplings from mass to weak basis
`getQuarkMixedYukawas`	sets all entries of quark Yukawa couplings given fermion masses, Higgs VEV and CKM matrix

Table 9. `MssmSusy` class.

E.8 `MssmSoftsusy` Class

`MssmSoftSusy` objects contain a structure `sPhysical` encapsulating the physical information on the superparticles (pole masses and physical mixings), as shown in table 11. Another structure of type `drBarPars` inherits from `sPhysical` but instead contains information on $\overline{DR}$ masses and mixing angles. `MssmSoftSusy` objects also contain one of these structures for calculational convenience: the information is used in order to calculate loop corrections to various masses. As table 12 shows, a method `mpzCharginos` returns the 2 by 2 complex diagonalisation matrices U, V that result in positive $\overline{DR}$ chargino masses, as defined in ref. [3]. The method `mpzNeutralinos` is present in order to convert O to the complex matrix N defined in ref. [3] that would produce only positive $\overline{DR}$ neutralino masses. This information, as well as $\overline{DR}$ third family fermion masses are stored in the `drBarPars` structure.

`MssmSoftsusy` data variables and accessors can be viewed in table 14 and the most important high-level methods are displayed in table 15. The operators `<<`, `>>` have been overloaded to write or read `MssmSoftusy` objects or `sPhysical` structures to/from a file stream. The driver routine for the RGE evolution and unification calculation is

data		methods
`double m32` $m_{3/2}$	compensator VEV* (GeV)	`setM32` `displayGravitino`
`DoubleVector mGaugino` $M_{1,2,3}$	(1 . . . 3) vector of gaugino mass parameters	`setGauginoMass` `displayGaugino`

`DoubleMatrix ua,da,ea`
U_A, D_A, E_A
	(3 by 3) matrix of trilinear soft terms (GeV)	`setTrilinearElement`
		`displayTrilinearElement`
		`displaySoftA`

`DoubleMatrix mQLsq`
`mURsq,mDRsq,mLLsq`
`mSEsq`
$(m^2_{\tilde{Q}_L})$, $(m^2_{\tilde{u}_R})$, $(m^2_{\tilde{d}_R})$,
$(m^2_{\tilde{L}_L})$, $(m^2_{\tilde{e}_R})$
	(3 by 3) matrices of soft SUSY breaking masses (GeV^2)	`setSoftMassElement`
		`setSoftMassMatrix`
		`displaySoftMassSquared`

`double m3sq,mH1sq,mH2sq`
m^2_3, $m^2_{H_1}$, $m^2_{H_2}$
	Bilinear Higgs parameters (GeV, GeV^2, GeV^2)	`setM3Squared`
		`setMh1Squared`
		`setMh2Squared`
		`displayM3Squared`
		`displayMh1Squared`
		`displayMh2Squared`

method	function
`standardSugra`	Sets all universal soft terms
`universalScalars`	Sets universal scalar masses
`universalGauginos`	Sets universal gaugino masses
`universalTrilinears`	Sets universal soft breaking trilinear couplings
`addAmsb`	Adds AMSB soft terms to current object*
`minimalGmsb`	Gauge-mediated soft terms used as boundary conditions*

Table 10. `SoftParsMssm` class data and methods. The asterisk denotes additional information in the text.

```
double MssmSoftsusy::lowOrg
(void (*boundaryCondition)(MssmSoftsusy &, const DoubleVector &),
 double mxGuess, const DoubleVector & pars, int sgnMu, double tanb, const QedQcd &
 oneset, bool gaugeUnification, bool ewsbBCscale = false)
```

The user-supplied `boundaryCondition` function sets the soft parameters according to the elements of the supplied `DoubleVector` at `mxGuess`, as discussed in appendix B. If `gaugeUnification` is `true`, the scale that unifies the electroweak gauge couplings is used and returned by the function. If `gaugeUnification` is `false`, the function simply returns `mxGuess`. `ewsbBCscale` is an optional argument: if one wishes to impose the soft SUSY breaking boundary conditions at M_{SUSY} rather than M_X, one should call `lowOrg` with the value `true` for this argument. Omitting the argument, or giving it the default `false` value means that M_X will be used instead. `pars` contains a `DoubleVector` of soft SUSY breaking parameters to be applied as the theoretical boundary condition. `sgnMu` is the sign of the superpotential μ parameter, `tanb` is the value of $\tan\beta(M_Z)$ required and `oneset` contains the $M_{QED\times QCD5}$ scale low

data variable	description
`double mh0,mA0,mH0,mHpm`	$h^0, A^0, H^0, H^{\pm}$masses
`DoubleVector msnu`	vector of $m_{\tilde{\nu}_{i=1\ldots3}}$ masses
`DoubleVector mch,mneut`	vectors of $m_{\chi^{\pm}{}_{i=1\ldots2}}$, $m_{\chi^0{}_{i=1\ldots4}}$ respectively
`double mGluino`	gluino mass $m_{\tilde{g}}$
`DoubleMatrix mixNeut`	4 by 4 orthogonal neutralino mixing matrix O
`double thetaL, thetaR`	$\theta_{L,R}$ chargino mixing angles
`double thetat, thetab`	$\theta_{t,b}$ sparticle mixing angles
`double thetatau, thetaH`	θ_{τ}, α sparticle and Higgs mixing angles
`DoubleMatrix mu, md, me`	(2 by 3) matrices of up squark, down squark and charged slepton masses

Table 11. `sPhysical` structure. Masses are pole masses, and stored in units of GeV. Mixing angles are in radian units.

data variable	description
`double mt, mb, mtau` $m_t(Q), m_b(Q), m_{\tau}(Q)$	Third family fermion masses
`DoubleVector mnBpmz, mchBpmz` $m_{\chi_i^0}, m_{\chi_i^{\pm}}$	Positive neutralino and chargino masses $(1\ldots4,\ 1\ldots2)$ vectors
`ComplexMatrix nBpmz` N	Neutralino mixing matrix (4 by 4 complex matrix)
`ComplexMatrix uBpmz, vBpmz` U, V	Chargino mixing matrices (2 by 2 complex matrices)

name	function
`mpzNeutralino`	Gives mixing matrices required to make neutralino masses positive*
`mpzChargino`	Gives mixing matrices required to make chargino masses positive*

Table 12. `drBarPars` structure. Masses are in the $\overline{DR}$ scheme, and stored in units of GeV. Mixing angles are in radian units. Functions marked with an asterisk are mentioned in the text.

energy data. $M_{QED\times QCD5}$ is defined to be the scale of `oneset`. If `altEwsb=true`, we have alternative boundary conditions on electroweak symmetry breaking, and the data in the class `AltEwsbMssm` is used. `qewsb` may be used to multiplicatively change the scale M_{SUSY} at which the Higgs potential is minimised and sparticle masses calculated, or it may alternatively be used as a fixed scale. This can be useful if one wants to examine the scale dependence of the results [31]. Setting `qewsb`=$x < M_Z/1$ GeV, sets $M_{SUSY} = x\sqrt{m_{\tilde{t}_1}(M_{SUSY})m_{\tilde{t}_2}(M_{SUSY})}$. Values from 0.5 to 2 are common. If $M_{SUSY} < M_Z$ results from the above expression, $M_{SUSY} = M_Z$ is used. If on the other hand a *fixed scale* is desired from M_{SUSY}, setting `qewsb`$> M_Z/1$ GeV in the input results in M_{SUSY} =`qewsb` GeV being fixed.

This class has just two private data variables, as shown in Table 16, where both are massive parameters in units of GeV. They fix the boundary conditions on $\mu(M_{SUSY})$ and the pole pseudo-

data variable	flags
`mgutOutOfBounds`	$M_X > 5 \times 10^{17}$ GeV or $M_X < 10^4$ GeV
`irqfp`	in a region with a Landau pole
`noRhoConvergence`	the ρ iterative routine doesn't converge
`noConvergence`	the main iteration routine doesn't converge
`tachyon`	a non-Higgs scalar has negative mass squared
`tachyonWarning`	a tree-level non-higgs imaginary mass at M_Z
`muSqWrongSign`	μ^2 from eq. (3.11) negative
`m3sq`	m_3^2 from eq. (3.12) has incorrect sign
`higgsUfb`	eq. (C.1) is not satisfied
`nonperturbative`	a Landau pole was reached below the unification scale
`noMuConvergence`	μ could not be calculated reliably
`inaccurateHiggsMass`	Higgs masses cannot be trusted
`problemThrown`	numerical exception occurred during run (infinities etc)

Table 13. `sProblem` structure. All data variables are boolean values except for `tachyon`, which is of type `tachyonType` and `tachyonWarning`. See Section C.1 for more details.

scalar Higgs mass m_A(pole). In this case, $m_A(M_{SUSY})$ is extracted from the input m_A(pole) via $m_A^2(M_{SUSY}) = m_A^2(\text{pole}) + \Pi_{AA}(M_{SUSY})$, where $\Pi_{AA}(M_{SUSY})$ is the MSSM pseudo-scalar self-energy correction. The Higgs mass squared parameters $m_{H_{1,2}}^2$ are *not* set at M_{GUT}: they are instead set at M_{SUSY} by solving the simultaneous equations 3.11, 3.12 and using the relationship between $m_3^2(M_{SUSY})$ and $m_A(M_{SUSY})$:

$$\begin{aligned} m_{H_1}^2 &= \sin^2\beta(m_A^2 + M_Z^2) + \frac{t_1}{v_1}(1 - \sin^4\beta) - \sin^2\beta\cos^2\beta\frac{t_2}{v_2} - (\mu^2 + \frac{1}{2}M_Z^2) \\ m_{H_2}^2 &= \cos^2\beta(m_A^2 + M_Z^2) + \frac{t_2}{v_2}(1 - \cos^4\beta) - \sin^2\beta\cos^2\beta\frac{t_1}{v_1} - (\mu^2 + \frac{1}{2}M_Z^2), \end{aligned} \tag{E.3}$$

where all quantities in eq. E.3 are running parameters evaluated at M_{SUSY}. This option is covered under the SLHA input parameters `EXTPAR 23,26` [30].

The fine tuning (as defined in sec. 3) can be calculated with the method

```
DoubleVector MssmSoftsusy::fineTune(void (*boundaryCondition)
        (MssmSoftsusy &, const DoubleVector &), const DoubleVector
         & bcPars, double mx) const
```

This function should only be applied to an `MssmSoftsusy` object which has been processed by `lowOrg`. `mx` is the unification scale and `boundaryCondition` is the function that sets the unification scale soft parameters, as discussed above. In derived objects, the virtual method `methodBoundaryCondition` may be used to set data additional to `MssmSoftsusy` from the `boundaryCondition` function. The method outputs the fine-tuning of a parameter $a_{i=1\ldots n}$ in the `bcPars(n+3) DoubleVector`, with the $(n+1, n+2, n+3)^{th}$ element of `bcPars` being the fine-tuning with respect to the Higgs potential parameters (μ and B) and the top Yukawa coupling (h_t) respectively. `fineTune` is an optional feature. Alternatively, for a Bayesian measure of naturalness of the soft parameters only, one may use the method `double calcBayesianNaturalness() const` [35]. This method returns `nan` in the case of numerical instabilities in the naturalness calculation. `sinSqThetaEff()` returns a `double` number corresponding to a full one-

data		methods
`double mw`	pole M_W prediction	`setMw`
M_W	(GeV)	`displayMw`
`double msusy`	Minimisation scale	`displayMsusy`
M_S	(GeV)	`setMsusy`
`QedQcd dataset`	M_Z boundary condition on	`setData`
	Standard Model couplings	`displayDataSet`
`sProblem problem`	problem flags	`displayProblem`
		`flagIrqfp`, `flagB`
		`flagNonperturbative`
		`flagTachyon`, `flagHiggsufb`
		`flagNoConvergence`
		`flagNoMuConvergence`
		`flagNoRhoConvergence`
		`flagMusqwrongsign`
		`flagAllProblems`
`DrBarPars forLoops`	$\overline{DR}$ masses and mixings	`displayDrBarPars`
		`setDrBarPars`
`sPhysical physpars`	pole masses and mixings	`displayPhys`
		`setPhys`
`bool setTbAtMX`	flag: `true` if $\tan\beta$ is fixed	`displaySetTbAtMX`
	at M_{SUSY}	`setTbAtMX`
`double fracDiff`	fractional difference to last iteration	`displayFracDiff`
`bool altEwsb`	flag: `true` for alternative EWSB	`displayAltEwsb`
	conditions	
`double mxBC`	Boundary condition scale for	`displayMxBC`
	setting SUSY breaking terms (GeV)	`setMxBC`
`double t1OV1Ms`	2-loop tadpole $t_1/v_1(M_{SUSY})$	`displayTadpole1Ms`
`double t2OV2Ms`	2-loop tadpole $t_2/v_2(M_{SUSY})$	`displayTadpole2Ms`
`double t1OV1Ms1loop`	1-loop tadpole $t_1/v_1(M_{SUSY})$	`displayTadpole1Ms1loop`
`double t2OV2Ms1loop`	1-loop tadpole $t_2/v_2(M_{SUSY})$	`displayTadpole2Ms1loop`
`double qewsb`	User-set EWSB scale (GeV)	`displayQewsb`
	default `1.0`	`setQewsb`

Table 14. `MssmSoftsusy` class data and accessor methods.

loop calculation of the quantity $\sin^2\theta^l_{eff}$. It does *not* contain any 2-loop corrections, and may not be accurate enough for precision electroweak fits.

E.9 `FlavourMssmSoftsusy` Class

`FlavourMssmSoftsusy` objects inherit directly from `MssmSoftsusy` with the addition of the variables shown in Table 17. Some of these constitute the angles of the PMNS and CKM matrices in the standard parameterisation of eq. 3.7. The `lesHouchesAccordOutput` method has been overloaded to provide flavour violating input and output in accordance with the SLHA2 conventions [43]. Note that `SOFTSUSY` currently does *not* contain CP-violating complex phases, despite the inclusion of δ. Thus,

name	function
`lowOrg`	Driver routine for whole calculation*
`methodBoundaryCondition`	Boundary condition for derived objects*
`itLowsoft`	Performs the iteration between M_Z and unification scale
`sparticleThresholdCorrections`	$\overline{DR}$ radiative corrections to Standard Model couplings at M_Z
`physical`	Calculates sparticle pole masses and mixings
`calcDrBarPars`	Calculates $\overline{DR}$ pole masses and mixings
`rewsb`	Sets μ, B from EWSB conditions
`fineTune`	Calculates fine-tuning for soft parameters* and h_t
`calcBayesianNaturalness`	Calculates Bayesian naturalness for soft parameters
`getVev`	Calculates VEV $v^{\overline{DR}}$ at current scale from Z self-energy and gauge couplings
`calcSinthdrbar`	Calculates $s_W^{\overline{DR}}$ at current scale from gauge couplings
`calcMs`	Calculates M_{SUSY}
`printShort`	short list of important parameters printed out to standard output in columns
`printLong`	long list of important parameters printed out to standard output in columns
`outputFcncs`	prints a list of flavour-changing δ parameters*
`lesHouchesAccordOutput`	prints output in SLHA [30] format
`sinSqThetaEff`	calculates then returns $\sin^2\theta^l_{eff}$
`useAlternativeEwsb`	switches `altEwsb=true`, using alternative EWSB

Table 15. `MssmSoftsusy` methods and related functions. Functions marked with an asterisk are mentioned in the text.

data		methods
`muCond`	Alternative electroweak boundary	`displayMuCond`
$\mu(M_{SUSY})$	conditions	`setMuCond`
`maCond`	Alternative electroweak boundary	`displayMaCond`
M_A(pole)	conditions	`setMaCond`

Table 16. `AltEwsbMssm` class data and accessor methods.

when CKM angles are input via the SLHA2 in the Wolfenstein parameterisation, the magnitude of the 13 entry is fit to $\sin\theta_{13}$ as in eq. 3.8 and δ is set to zero (for now). The relevant method is

```
void FlavourMssmSoftsusy::setAngles
(double lambdaW, double aCkm, double rhobar, double etabar)
```

Currently, an identical parameterisation is used for the PMNS matrix that describes lepton mixing, except θ_{ij} is replaced by $\bar{\theta}_{ij}$. There is currently no provision for a CP-violating phase. Effective light

data		methods
`flavourPhysical fv`	sfermion mixing/mass data	`setFlavourPhys`
		`displayFlavourPhys`
`double theta12, theta23`	CKM matrix parameters	`setTheta12`
`double theta13, deltaCkm`		`setTheta13`
$\theta_{12}, \theta_{23}, \theta_{13}, \delta$		`setTheta23`
		`setDelta`
		`displayTheta12`
		`displayTheta23`
		`displayTheta13`
		`displayDelta`
`double thetaB12, thetaB23`	PMNS matrix parameters	`setThetaB12`
`double thetaB13`		`setThetaB13`
$\bar{\theta}_{12}, \bar{\theta}_{13}, \bar{\theta}_{23}$		`setThetaB23`
		`displayThetaB12`
		`displayThetaB23`
		`displayThetaB13`
`double mNuE, mNuMu, mNuTau`	light neutrino masses	`setMnuE`
$m_{\nu_e}, m_{\nu_\mu}, m_{\nu_\tau}$		`setMnuNu`
		`setMnuTau`
		`displayMnuE`
		`displayMnuMu`
		`displayMnuTau`
`double mcMc`	running charm quark mass	`setMcMc`
$m_c(m_c)$		`displayMcMc`
`double md2GeV, mu2GeV, ms2GeV`	quark masses at 2 GeV	`setMd2GeV`, `setMu2GeV`
$m_d(2), m_u(2), m_s(2)$		`setMs2GeV`
		`displayMd2GeV`
		`displayMu2GeV`
		`displayMs2GeV`
`double mePole, mmuPole`	pole lepton masses	`setPoleMe, setPoleMmu`
m_e^{pole}, m_μ^{pole}		`displayPoleMe`
		`displayPoleMmu`

Table 17. `FlavourMssmSoftsusy` class data and accessor methods. All angles are measured in radians and masses are measured in GeV.

neutrino masses are also included in the object.

An optional feature intended for studies of flavour-changing neutral currents (FCNCs) is the method

```
void FlavourMssmSoftsusy::sCkm
(DoubleMatrix & deltaULL, DoubleMatrix & deltaURR, DoubleMatrix & deltaULR,
 DoubleMatrix & deltaDLL, DoubleMatrix & deltaDRR, DoubleMatrix & deltaDLR)
const
```

data	description
`DoubleMatrix dSqMix`	6×6 down squark mixing
`DoubleMatrix uSqMix`	6×6 up squark mixing
`DoubleVector msD`	6 mass-ordered down squark masses
`DoubleVector msU`	6 mass-ordered up squark masses
`DoubleMatrix eSqMix`	6×6 charged slepton mixing matrix
`DoubleMatrix nuSqMix`	3×3 sneutrino mixing matrix
`DoubleVector msE`	6 mass-ordered charged slepton masses
`DoubleVector msNu`	3 mass-ordered sneutrino masses

Table 18. `flavourPhysical` structure. All mass parameters are pole masses, stored in units of GeV. The mixing matrix definitions exactly coincide with those in the SLHA2 [43] and describe the transformation between the loop-corrected super CKM basis and the mass basis.

which calculates the parameters $(\delta^{u,d}_{LL,LR,RR})_{ij}$ calculated in the mass-squared-insertion approximation (after a rotation to the super CKM basis), as defined in Ref. [10].

The relationship between the super CKM basis of the mass matrices (including one-loop corrections) and the pole-mass basis as described by SLHA2, is contained in the structure `flavourPhysical`. We list the relevant data in Table 18. Note that, in the case that CMSSM type inputs are listed in the SLHA2 *as well as* the flavour input parameters $\hat{T}_{U,D,E}$ (assumed to be in the superCKM basis), a special procedure is required to implement both types of term. At M_X, the A_0 terms are added in the interaction basis. There is then a transformation to the super CKM basis, where any values of $\hat{T}_{U,D,E}$ input over-write the trilinear terms. Finally, the trilinear terms are transformed back to the interaction basis.

Acknowledgments

This work has been partially supported by PPARC, STFC grant ST/P000681/1, the KUTs workshops and the Aspen Center for Physics. We would like to thank A. Djouadi, S. Kraml and W. Porod for help with detailed comparisons [9] with `SUSPECT`, `ISAJET` and `SPHENO`, S. Akula, P. Athron, M. Baak, M. Badziak, E. Bagnaschi, G. Bélanger, F. Borzumati, F. Boudjema, F. Brummer, A. Buckley, M. Catalan, A. Casas, T. Cridge, M. Eads, H. Eberl, J. Evans, R. Franceschini, R. Harlander, M. Hamer, M. Hussain, J. Hetherington, M. Ibe, A. Ibrahim, M. Iskrzynski, J. Kersten, J. Klappert, S. Knapen, S. Kom, S. Krippendorf, S. Kulkarni, D. Kunz, K. Ishikawa, R. Lu, N. Mahmoudi, S. Martin, P. Meade, P. Millet, D. Morrissey, L. Nicoletta, B. Petersen, A. Pukhov, M. Ramage, R. Ruiz de Austri, S. Sekmen, A. Sheplyakov, D. Shih, P. Slavich, P. Skands, J. Smidt, F. Staub, C. Tamarit, S. Tetsuo, L. Tunstall, A. Voigt, T. Watari and A. Wingerter for bug finding and coding suggestions, I. Gafton, J. Holt, F. Krauss, D. Sanford, C. Wymant and F. Yu for suggestions on the draft, D. Ross for information on the Passarino-Veltman integrals, K. Matchev for other useful discussions and G. de Grassi and G. Pier Paolo for providing the fit to the higher-order corrections to electroweak gauge couplings.

References

[1] G. G. Ross and R. G. Roberts, *Minimal supersymmetric unification predictions*, *Nucl. Phys.* **B377** (1992) 571–592.

[2] V. Barger, M. S. Berger, and P. Ohmann, *The supersymmetric particle spectrum*, *Phys. Rev.* **D49** (1994) 4908–4930, [hep-ph/9311269].

[3] D. M. Pierce, J. A. Bagger, K. Matchev, and R. jie Zhang, *Precision corrections in the minimal supersymmetric standard model*, *Nucl. Phys.* **B491** (1997) 3–67, [hep-ph/9606211].

[4] J. K. A. Djouadi and G. Moultaka, A. Djouadi, J. -L. Kneur and G. Moultaka, *SuSpect: A Fortran code for the supersymmetric and Higgs particle spectrum in the MSSM*, Comput. Phys. Commun. **176** (2007) 426 [hep-ph/0211331].

[5] H. Baer, F. E. Paige, S. D. Protopopescu, and X. Tata, *Isajet 7.48: A monte carlo event generator for p p, anti-p p, and e+ e- reactions*, hep-ph/0001086.

[6] B. C. Allanach, J. P. J. Hetherington, M. A. Parker, and B. R. Webber, *Naturalness reach of the large hadron collider in minimal supergravity*, *JHEP* **08** (2000) 017, [hep-ph/0005186].

[7] G. Corcella *et. al.*, *Herwig 6: An event generator for hadron emission reactions with interfering gluons (including supersymmetric processes)*, *JHEP* **01** (2001) 010, see the HERWIG web page http://hepwww.rl.ac.uk/theory/seymour/herwig/, [hep-ph/0011363].

[8] B. C. Allanach, *SUSY Predictions and SUSY Tools at the LHC*, Eur. Phys. J. **59** (2009) 427, arXiv:0805.2088 [hep-ph].

[9] B. C. Allanach, *Theoretical Uncertainties in Sparticle Mass Predictions*, Snowmass Proceedings 2001, [hep-ph/0110227]; B. C. Allanach *et al. The Beyond the Standard Model Working Group: Summary Report*, Les Houches proceedings 2002, [hep-ph/0204031]. B. C. Allanach, S. Kraml and W. Porod, *Comparison of SUSY mass spectrum calculations*, SUSY 2002 proceedings, [hep-ph/0207314]; B. C. Allanach, S. Kraml and W. Porod, *Theoretical uncertainties in sparticle mass predictions from computational tools*, JHEP **0303** (2003) 016, hep-ph/0302102; B. C. Allanach, G. Bélanger, F. Boudjema, A. Pukhov and W. Porod, *Uncertainties in Relic Density Calculations in mSUGRA*, hep-ph/0402161; *ibid* in *Beyond the Standard Model Working Group: Summary Report*, Les Houches "Physics at TeV Colliders 2003", hep-ph/0402295.

[10] F. Gabbiani, E. Gabrielli, A. Masiero, and L. Silvestrini, *A complete analysis of fcnc and cp constraints in general susy extensions of the standard model*, *Nucl. Phys.* **B477** (1996) 321–352, [hep-ph/9604387].

[11] S. G. Gorishnii, A. L. Kataev, S. A. Larin, and L. R. Surguladze, *Corrected three loop qcd correction to the correlator of the quark scalar currents and gamma (tot) (h0 → hadrons)*, *Mod. Phys. Lett.* **A5** (1990) 2703–2712.

[12] O. V. Tarasov, A. A. Vladimirov, and A. Y. Zharkov, *The gell-mann-low function of qcd in the three loop approximation*, *Phys. Lett.* **B93** (1980) 429–432.

[13] S. G. Gorishnii, A. L. Kataev, and S. A. Larin, *Next next-to-leading perturbative qcd corrections and light quark masses*, *Phys. Lett.* **B135** (1984) 457.

[14] K. G. Chetyrkin, B. A. Kniehl, and M. Steinhauser, *Strong coupling constant with flavour thresholds at four loops in the ms-bar scheme*, *Phys. Rev. Lett.* **79** (1997) 2184–2187, [hep-ph/9706430].

[15] L. V. Avdeev and M. Y. Kalmykov, *Pole masses of quarks in dimensional reduction*, *Nucl. Phys.* **B502** (1997) 419, [hep-ph/9701308].

[16] A. Bednyakov, A. Onishchenko, V. Velizhanin and O. Veretin, *Two-loop $O(\alpha_s^2)$ MSSM corrections to the pole masses of heavy quarks*, Eur. Phys. J. C **29** (2003) 87, [hep-ph/0210258];

[17] **Particle Data Group** Collaboration, see the latest Review of Particle Physics on [http://pdg.lbl.gov/] D. E. Groom *et. al.*, *Review of particle physics*, *Eur. Phys. J.* **C15** (2000) 1.

[18] G. Degrassi, P. Gambino and P. P. Giardino, *The $m_W - m_Z$ interdependence in the Standard Model: a new scrutiny*, [arXiv:1411.7040 [hep-ph]].

[19] J. A. Casas, J. R. Espinosa, and H. E. Haber, *The higgs mass in the mssm infrared fixed point scenario*, *Nucl. Phys.* **B526** (1998) 3–20, [hep-ph/9801365].

[20] S. P. Martin, *Two-loop effective potential for a general renormalizable theory and softly broken supersymmetry*, *Phys. Rev.* **D65** (2002) 116003 [hep-ph/0111209].

[21] Y. Yamada, *Two-loop renormalization of tan(beta) and its gauge dependence*, *Phys. Lett.* **B530** (2002) 174 [hep-ph/0112251].

[22] A. Dedes and P. Slavich, *Two-loop corrections to radiative electroweak symmetry breaking in the mssm*, Nucl. Phys. B **657** (2003) 333, hep-ph/0212132.

[23] A. Katsikatsou, A. B. Lahanas, D. V. Nanopoulos, and V. C. Spanos, *On the radiative corrections to the pseudo-scalar higgs boson mass*, *Phys. Lett.* **B501** (2001) 69–77, [hep-ph/0011370].

[24] G. Degrassi, P. Slavich, and F. Zwirner, *On the neutral higgs boson masses in the mssm for arbitrary stop mixing*, *Nucl. Phys.* **B611** (2001) 403–422, [hep-ph/0105096].

[25] A. Brignole, G. Degrassi, P. Slavich, and F. Zwirner, *On the o(alpha**2(t)) two-loop corrections to the neutral higgs boson masses in the mssm*, *Nucl. Phys.* **B631** (2002) 195–218, [hep-ph/0112177].

[26] A. Brignole, G. Degrassi, P. Slavich, and F. Zwirner, *On the two-loop sbottom corrections to the neutral higgs boson masses in the mssm*, *Nucl. Phys.* **B643** (2002) 79–92, [hep-ph/0206101].

[27] A. Dedes, G. Degrassi and P. Slavich, *On the two-loop Yukawa corrections to the MSSM Higgs boson masses at large tan(beta)*, Nucl. Phys. B **672** (2003) 144, hep-ph/0305127.

[28] R. Barbieri and A. Strumia, *About the fine-tuning price of lep*, *Phys. Lett.* **B433** (1998) 63–66,[hep-ph/9801353].

[29] B. de Carlos and J. A. Casas, *One loop analysis of the electroweak breaking in supersymmetric models and the fine tuning problem*, *Phys. Lett.* **B309** (1993) 320–328, [hep-ph/9303291].

[30] P. Skands *et al*, *Interfacing SUSY Spectrum Calculators, Decay Packages and Event Generators*, *JHEP* **0407** (2004) 036, [hep-ph/0311123].

[31] B. C. Allanach, A. Djouadi, J. L. Kneur, W. Porod and P. Slavich, *Precise determination of the neutral Higgs boson masses in the MSSM*, JHEP **0409** (2004) 044, arXiv:hep-ph/0406166; B. C. Allanach, G. A. Blair, S. Kraml, H. U. Martyn, G. Polesello, W. Porod and P. M. Zerwas, *Reconstructing supersymmetric theories by coherent LHC / LC analyses*, arXiv:hep-ph/0406166.

[32] S. P. Martin and M. T. Vaughn, *Two Loop Renormalization Group Equations For Soft Supersymmetry Breaking Couplings*, Phys. Rev. D **50** (1994) 2282 [arXiv:hep-ph/9311340].

[33] H. Baer, J. Ferrandis, K. Melnikov and X. Tata, *Relating bottom quark mass in $\overline{DR}$ and $\overline{MS}$ schemes*, *Phys. Rev.* **D66** (2002) 074007, [hep-ph/0207126].

[34] S. S. AbdusSalam, B. C. Allanach, H. K. Dreiner, J. Ellis, U. Ellwanger, J. Gunion, S. Heinemeyer and M. Kraemer *et al.*, *Benchmark Models, Planes, Lines and Points for Future SUSY Searches at the LHC*, Eur. Phys. J. C **71** (2011) 1835 [arXiv:1109.3859 [hep-ph]].

[35] P. Athron, C. Balazs, B. Farmer, A. Fowlie, D. Harries and D. Kim, *Bayesian analysis and naturalness of (Next-to-)Minimal Supersymmetric Models*, JHEP **1710** (2017) 160, doi:10.1007/JHEP10(2017)160 [arXiv:1709.07895 [hep-ph]].

[36] B.C. Allanach *et al*, *The Snowmass Points and Slopes: Benchmarks for SUSY Searches*, *Eur. Phys. J.* **C25** (2002) 113, [hep-ph/0202233].

[37] M. Muhlleitner, A. Djouadi and Y. Mambrini, *SDECAY: A Fortran code for the decays of the supersymmetric particles in the MSSM* Comput. Phys. Commun. **168** (2005) 46 [hep-ph/0311167].

[38] T. Sjostrand, L. Lonnblad, S. Mrenna and P. Skands, *PYTHIA 6.3 physics and manual*, [hep-ph/0308153].

[39] G. Belanger, F. Boudjema, A. Pukhov and A. Semikov, *micrOMEGAs: A program for calculating the relic density in the MSSM*, *Comput. Phys. Comm.* **149** (2002) 103, [hep-ph/0112278].

[40] B.C. Allanach and A. Voigt, *Uncertainties in the Lightest CP Even Higgs Boson Mass Prediction in the Minimal Supersymmetric Standard Model: Fixed Order Calculations Versus Effective Field Theory Predictions*, to appear.

[41] L. Randall and R. Sundrum, *Out of this world supersymmetry breaking*, *Nucl. Phys.* **B557** (1999) 79, [hep-th/9810155].

[42] S. Ambrosanio, G. D. Kribs and S. P. Martin, *Signals for gauge-mediated supersymmetry breaking models at the CERN LEP2 collider*, *Phys. Rev.* **D56** (1997) 1761 [hep-ph/9703211].

[43] B. C. Allanach, C. Balazs, G. Belanger, M. Bernhardt, F. Boudjema, D. Choudhury, K. Desch and U. Ellwanger *et al.*, *SUSY Les Houches Accord 2*, Comput. Phys. Commun. **180** (2009) 8 [arXiv:0801.0045 [hep-ph]].